



Review of micro seepage mechanisms in shale gas reservoirs

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ABSTRACT

Accurate understanding of gas micro seepage mechanisms in shale plays is of great importance for numerical simulation and productivity prediction. Classical seepage theory fails to build the constitutive relation of fluid flow in shale matrix, while current models show discrepancies from one to another. In this review, current bulk phase gas transporting models in literatures are classified into eight types, where consistency and diversity among them are revealed, with three recommended for shale gas seepage modeling on different purposes. The unification contributes to understand the role of different seepage mechanisms and helps to eliminate the confusion of different roles that each mechanism may play during mathematical modeling and gas production. Different approaches of handling geo-mechanical effects, ad-/de-sorption, real gas effects and rarefaction effects are introduced and compared, with recommended methods given after comparison. Based on practical pore size distribution, in-situ pressure and temperature, five ways of calculating gas mean free paths are compared, after which the possible flow regimes in practical shale plays are divided according to Knudsen number. Various empirical Klinkenberg correction methods and slip boundary conditions are summarized and compared, showing that some are inappropriate to describe rarefaction effects. Different ways of modeling Knudsen and surface diffusion are reviewed and analyzed. Considering the role of viscous flow, slippage effect, Knudsen diffusion and surface diffusion, gas transporting models are categorized into three types. The review provides the recent development and a systematic summarization on analytical modeling of gas transporting as well as the related unique phenomena in shale nanopores, which promotes the understanding of the complex and special micro seepage mechanisms in shale gas reservoirs.

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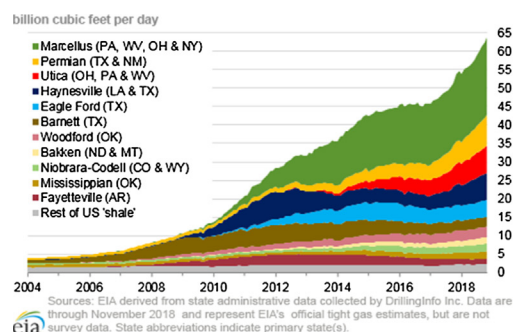
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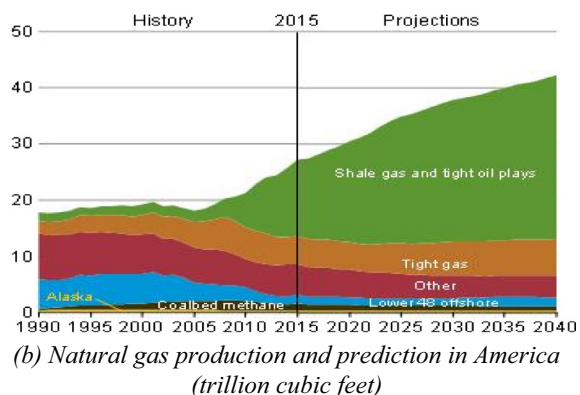
1. Introduction

Shale gas has become a hot research topic and potential long-term clean energy replacement due to its considerable reserves around the world and successful development in North America and China. According to the estimation of EIA (2018), shale gas has undergone significant production increase in the past decade from 2008 to 2018, as shown in Fig. 1(a). Meanwhile, Fig. 1(b) predicts that natural gas production from shale gas and tight oil plays leads growth in America natural gas supply from 2015 to 2040, accounting for about 55% contribution.

Large scale natural gas production from shale benefited from long horizontal well and mass hydraulic fracturing techniques. However, theoretical understanding of micro seepage mechanisms in shale matrix is still shallow and confusing, though great progress has been achieved in last few years. According to the updated status, molecular dynamics (MD) [1–3], lattice Boltzmann method (LBM) [4–6] and improved analytical models [7–9] are three main techniques to study micro seepage mechanisms in shale gas/oil reservoirs. The microscopic MD or mesoscopic LBM gets rid of the restriction of continuum media assumption and is capable of capturing gas micro seepage details. Yet, the computational cost



(a) Monthly dry shale gas production in America



(b) Natural gas production and prediction in America (trillion cubic feet)

Fig. 1. Current and predictive situations of shale gas production in America.

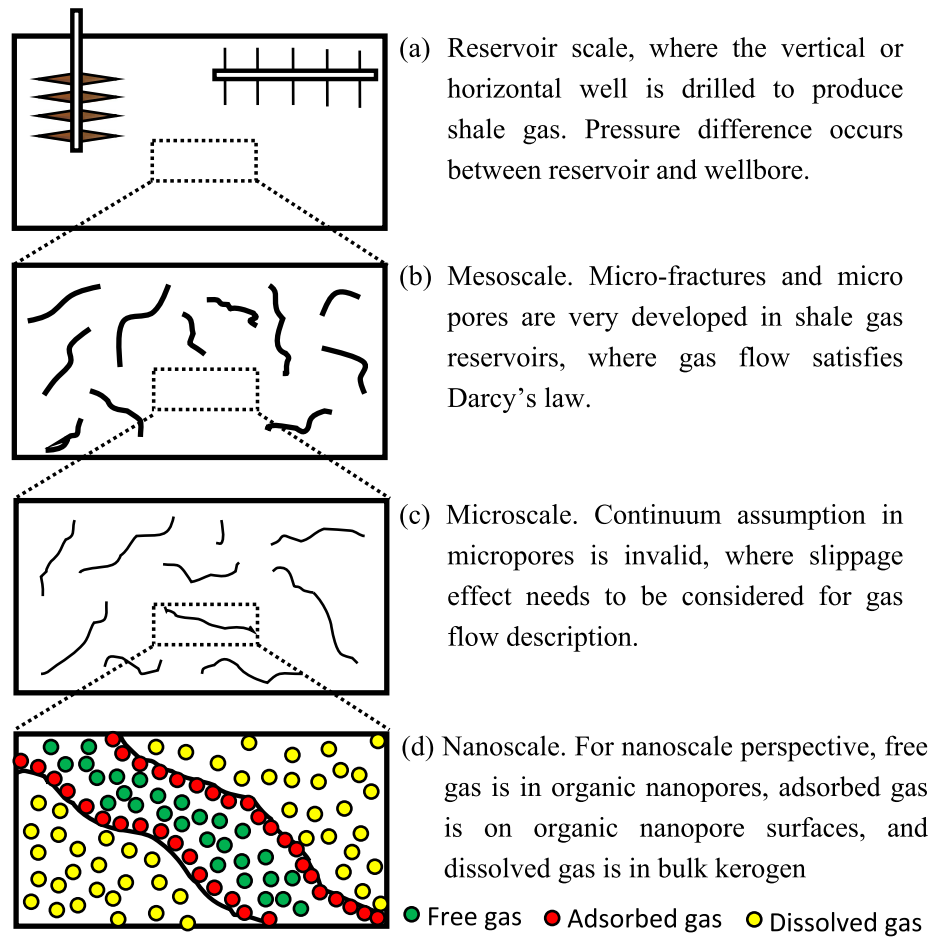


Fig. 2. Multi-scaled storage and flow paths in shale gas reservoirs as well as migration mechanisms (revised from Zhang et al. [20]).

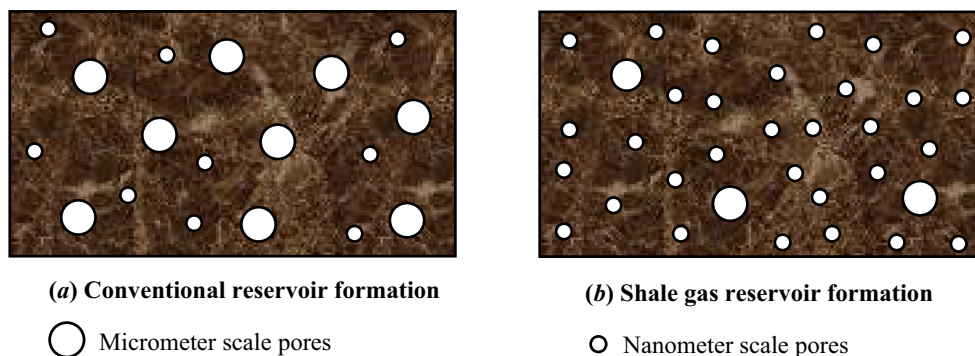


Fig. 3. Schematic of pore distribution in conventional reservoir (a) and shale gas reservoir (b), revised from Javadpour et al. [34].

is also huge for both techniques, which restricts its application into practical reservoir production. Instead, improved analytical models modify the continuum models and match the microscopic or mesoscopic results well if established appropriately. More importantly, it is an easy task to couple analytical models into numerical simulators, aiming at describing gas flow dynamics in unconventional reservoirs. Thus, accurate and reasonable analytical models are desirable for petroleum engineers to describe micro seepage behaviors in shale gas reservoirs.

Numerous analytical models [8–18] have been established to capture the unique gas transporting mechanisms in shale plays. Discrepancies and divergences could easily be found among them

by comparison. Some models employ Knudsen diffusion as a modification of slippage effect, while others claim the free molecular flow needs to be described by Knudsen diffusion. Upon the superposition of viscous flow, slippage effect and Knudsen diffusion, single slippage model and double slippage model could be classified according to the different role of Knudsen diffusion and slippage effect, which are quite different in nature. Various approaches were adopted to consider stress sensitivity effects, gas adsorption and desorption effects, real gas effects and rarefaction effects *et al.* in shale gas reservoirs, while quantitatively comparison and optimization of these methods are presently lacking. Generally, current research on gas micro seepage mechanisms in shale gas

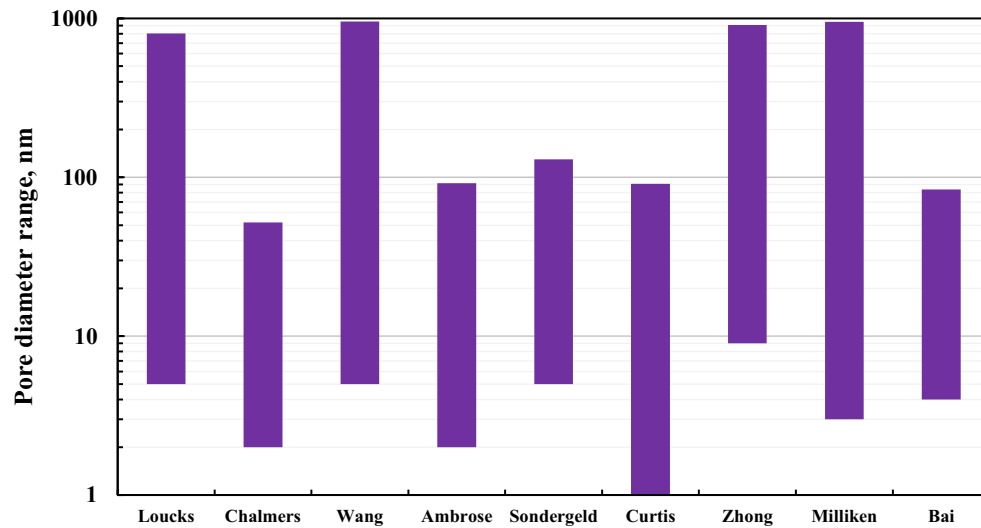


Fig. 4. Pore diameter ranges of organic matters from different studies [25–33].

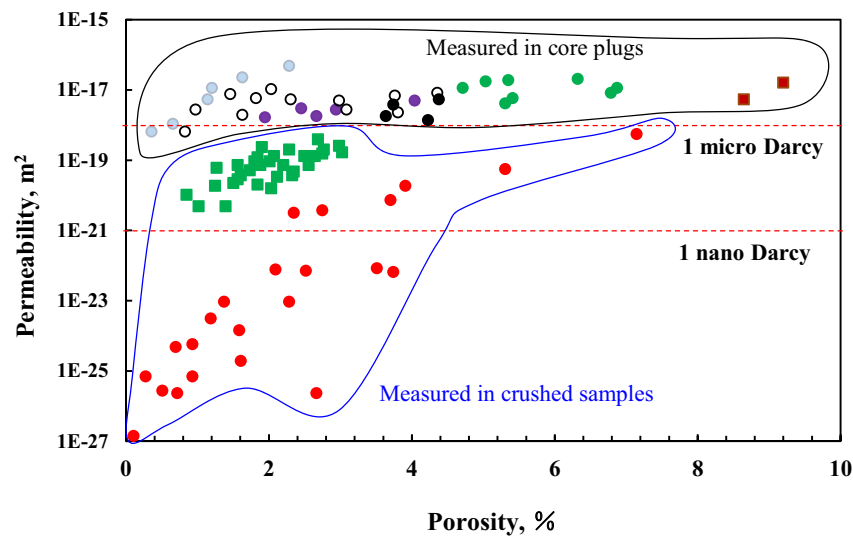


Fig. 5. Permeability and porosity relationships of shale samples from different shale gas reservoirs in north America [28,35,36].

reservoirs has drawn great attention, but fails to achieve a unified point of view, where the role of each influential factor or seepage mechanism is not fully clear.

In this review, our attention will mainly be paid on introducing the recent progress on analytical modeling of gas micro seepage mechanisms in shale gas nanopores as comprehensively as possible. The paper is arranged as follows: in Part 2 Gas transporting characteristics in shale nanopores, an overall gas transporting characteristics in shale gas reservoirs, including multi-scaled flow paths, pore size distribution, permeability and porosity ranges, are summarized; in Part3 Stress sensitivity effect, the commonly used models to account for permeability stress sensitivity effects are compared quantitatively; in Part4 Effects of adsorption and desorption on apparent permeability, different approaches of dealing with adsorption and desorption effects on permeability are summarized and compared; in Part 5 Gas deviation factor to account for real gas effects, approaches of obtaining gas deviation factors to account for real gas effects are summarized, where the calculation results were compared with digitalized Z-factor chart for method selection; in Part 6 Flow

chart and flow regimes division, different methods of calculating gas mean free path were summarized and compared, with possible flow regimes divided based on all above factors and practical conditions of shale gas reservoirs; in Part 7 Slippage effect, different Klinkenberg correction methods and various slip boundary conditions were collected from literatures and summarized; in Part 8 Knudsen diffusion, the Knudsen diffusion modeling for ideal gas and real gas was introduced respectively; in Part 9 Bulk phase gas transport models, two kinds of bulk phase gas transporting models were classified, where one is based on superposition method and the other is based on a function of Knudsen number; in Part 10 Model comparison, the different slip boundary conditions were compared first, and then various bulk phase gas transporting models were compared, with eight types of models classified and compared for unification; in Part 11 Surface diffusion, current surface diffusion models and their modeling procedures in shale gas plays are introduced and compared; in Part 12 Gas transporting model in shale nanopores, the gas transporting models based on bulk phase gas transport and surface diffusion are summarized into three types.

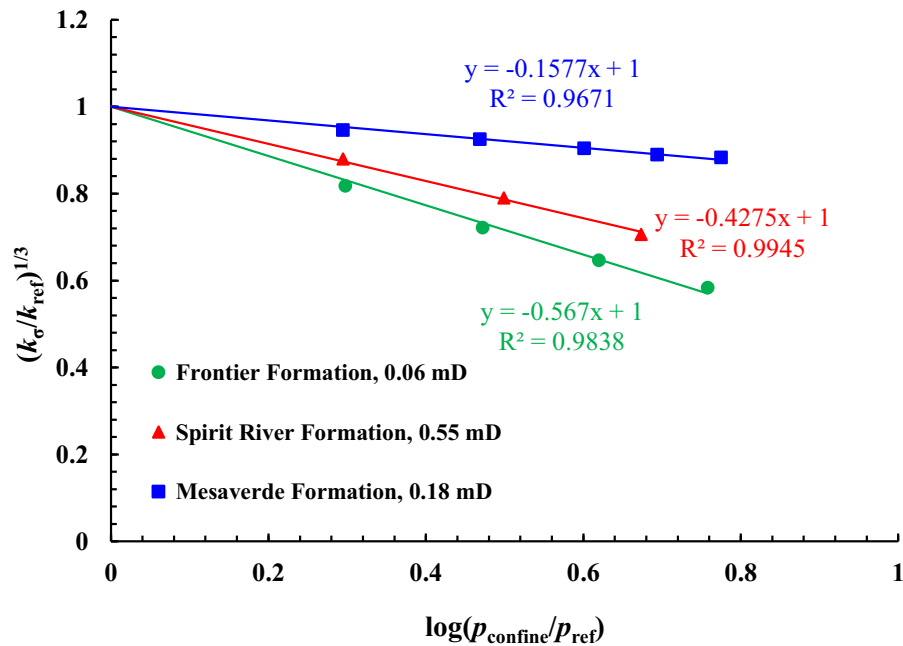


Fig. 6. The linear relationship between cubic root permeability and the logarithm of confining pressure (from Jones and Owens [37]).

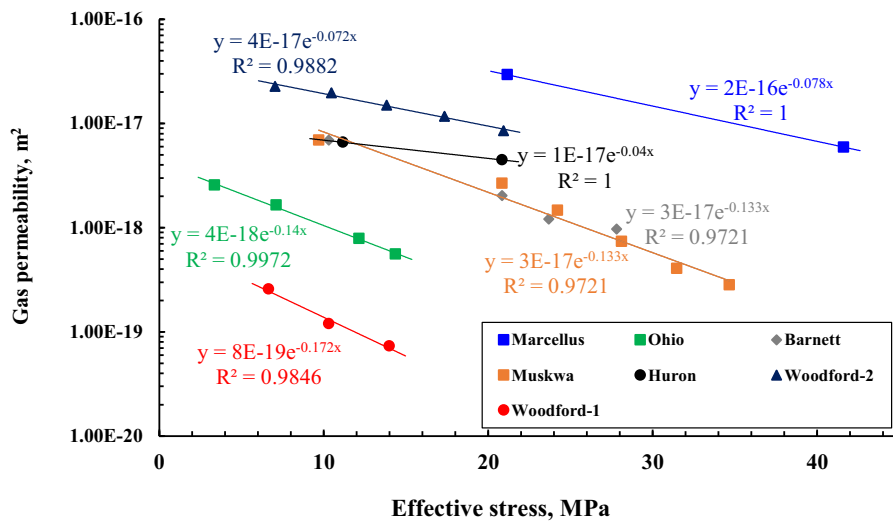


Fig. 7. The stress sensitivity of gas permeability in different shale gas reservoirs (changing confining pressure) [35,36,40].

2. Gas transporting characteristics in shale nanopores

Shale gas formation belongs to typical multi-scale porous media with multiple storage and migration mechanisms in it (Fig. 2), including possibly Darcy flow of free gas in hydraulic and natural fractures as well as in macropores, slippage effect and Knudsen diffusion of free gas in micropores, gas desorption from organic matter surfaces to bulk spaces [19], surface diffusion on organic pore surfaces and gas diffusion from organic kerogen to nanopores.

For gas transport in fractures or macro pores, the migration mechanism is clear and simple, i.e. gas flow obeys Darcy's law. Even for gas flow in micropores where continuum assumption might be invalid, it can still be characterized by Darcy's law with slip boundary conditions or considering Klinkenberg correction. However, gas transporting mechanisms in shale matrix system are still confusing and controversial due to the strong discontinuity.

Thus, in this review, gas flow behaviors in shale nanopores will be introduced in detail from mathematical modeling to model comparison for each migration mechanism, aiming to make clear the role of each flow mechanism in total gas transporting and give some insights into predicting shale gas flow dynamics more accurately.

Nanopores are very developed in shale gas reservoirs, comparing to conventional ones (Fig. 3), which provides large amount of surface area for gas adsorption and leads to unique gas migration characteristics. Adsorbed gas and dissolved gas are special gas storage forms in shale gas reservoirs, which act as supplements of free gas during gas production and prolong the production time of shale gas reservoirs [21]. Although it was reported that only free gas exists in inorganic ones [22], inorganic matter such as clay minerals may also provide much sites for gas adsorption [23,24]. Organic nanopores are very developed in shale gas reservoir [25–33], as can

be seen in Fig. 4, in which gas migration mechanisms are very different from those in inorganic nanopores. This is because gas adsorption in organic materials is mostly gas–solid interaction, while gas adsorption in inorganic pores needs to take the effects of initial water into account and belongs to gas–liquid–solid interaction as a result [24]. Therefore, shale gas migration behaviors in shale nanopores is quite complex and it is a very challenging task for petroleum engineers to mathematically model it with high accuracy.

Fig. 5 shows the relationships between permeability and porosity in different shale gas reservoirs in north America, where the data in blue circle are measured in crushed samples and data in black circle are measured in core plugs [36]. Permeability value measured in core plugs (mostly above 1 micro Darcy) are significantly higher than those measured in crushed samples (mostly below 1 micro Darcy), since the effects of natural and drilling induced microfractures on matrix permeability were eliminated. Therefore, true matrix permeability is recommended to be measured in crushed samples, while equivalent permeability of pores and microfracture networks should to be measured in core plugs. Basically, the values of porosity and permeability still satisfy linear relationship as in conventional oil or gas reservoirs. Generally, true matrix permeability of shale samples is below 1 micro Darcy, and even below 1 nano Darcy, and the porosity of shale samples is below 10% [19], which can also be seen from Fig. 5.

3. Stress sensitivity effect

As we know, the porosity and permeability changes with pressure depletion during production in many unconventional reservoirs. With the production of shale gas from reservoirs, the effective stress of formation increases continuously, which leads to a reduction in permeability due to the compaction of shale rocks. This may significantly reduce gas transporting capabilities and gas productivity, especially in stress sensitive reservoirs. Many studies have been done to couple the geo-mechanical effects into apparent permeability models to account for stress sensitivity of permeability in such cases, which is essential for accurate estimations of formation properties from well testing analysis and for developing accurate long-term production forecasts.

- (1) M-1: Based on experimental measurement, Jones and Owens [37] obtained a linear relationship between the cubic root of permeability versus the logarithm of confining pressure as below

$$k_{\sigma} = k_{\text{ref}} \left(1 - S_{\sigma} \log \frac{p_{\text{confining}}}{p_{\text{ref}}} \right)^3 \quad (1)$$

where k_{σ} is permeability considering effective stress; k_{ref} is permeability under reference confining pressure; p_{ref} is reference confining pressure; $p_{\text{confining}}$ is confining pressure; S_{σ} is a parameter denoting the sensitivity degree of confining pressure on permeability, which is also the slope of the regression line in Fig. 6. Increasing values of S_{σ} imply increasing effects of confining pressure on permeability. Generally, the lower the core permeability the more it is affected by confining pressure [37].

- (2) M-2: Another mathematical model that couples geo-mechanical and fluid flow aspects was also commonly employed to account for permeability change during production [15,38,39], which is

$$k_{\sigma} = k_{\text{ref}} e^{-\gamma(p_{\text{ref}} - p)} \quad (2)$$

where γ is a constant about stress dependent permeability; k_{ref} is permeability under reference pressure; combining Eq. (2) and Eq. (4), a linear relationship can easily be found between the logarithm of permeability and effective stress, as shown in Fig. 7.

Gas permeability of shale samples from different shale gas reservoirs decreases significantly with effective stress increasing, [35,36,40] which satisfies exponential function relationship with $R^2 > 0.97$ for all shale samples (Fig. 7), i.e. Eq. (2) describes the sensitivity of stress dependent permeability very well for various shale samples from North America. The stress dependent of permeability here is caused by the decrease of pore radius and irrelevant to seepage mechanisms change in pores, since effective stress change is achieved by changing confining pressure. One drawback of this experimental method is that the changing of confining pressure leads to a different lithostatic pressure comparing to original reservoir conditions.

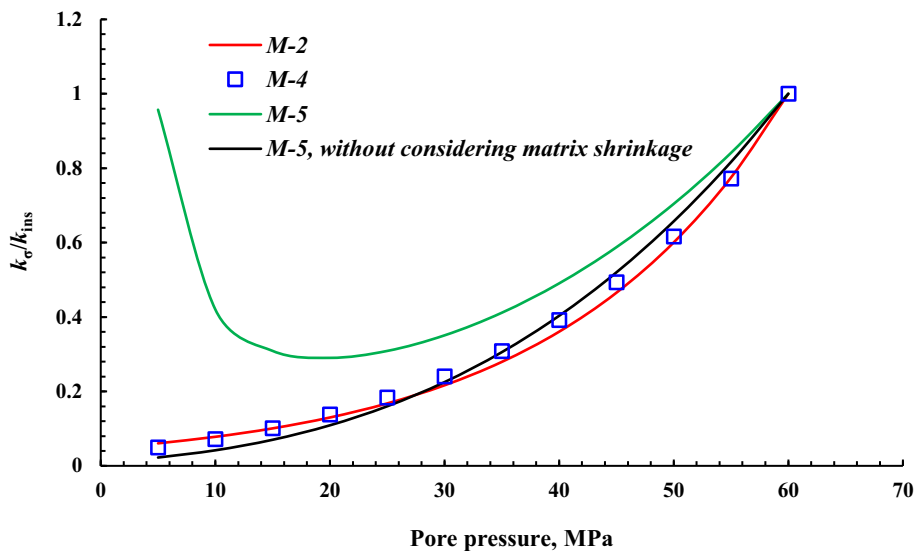


Fig. 8. The comparison of different permeability models for describing stress dependent.

- (3) M-3: Gangi's model [41] for fractured porous rock was also applied to describe stress dependence of apparent in some studies [22,42], though it was initially proposed to account for permeability variation of a fracture or fracture rock with confining pressure. This may due to the similar conditions between the two situations, where the shale nanopores is similar to that of small aperture in Gangi's model assumption. The model can be expressed as

$$k_{\sigma} = k_{ref} \left[1 - \left(\frac{\sigma_{eff}}{p_{mdls}} \right)^{m_g} \right]^3 \quad (3)$$

where σ_{eff} is effective stress; p_{mdls} is effective modulus of the asperities, or effective stress when pores are completely closed; m_g is a constant between 0 and 1. The effective stress can be calculated as

$$\sigma_{eff} = p_{confin} - \alpha_{stress} p \quad (4)$$

where α_{stress} is effective stress coefficient for permeability, which can be smaller or bigger than unity and determined by permeability sensitivity to confine pressure and pore pressure. Though effective stress coefficient α_{stress} is usually taken as a constant, it may be a function of pore pressure or confining pressure, which need to be further investigated by reliable experiments.

A more detailed review about effective stress as well as its influence on gas transporting processes can be seen in the study of Gensterblum et al. [43].

- (4) M-4: Some studies [12,44,45] considered the effects of stress dependence on rock porosity and permeability by the following power law, namely

$$\phi_{\sigma} = \phi_{atm} \left(\frac{\sigma_{eff}}{p_{atm}} \right)^{-n_n} \quad (5)$$

$$k_{\sigma} = k_{atm} \left(\frac{\sigma_{eff}}{p_{atm}} \right)^{-m_n} \quad (6)$$

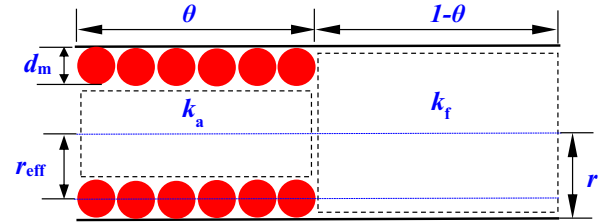


Fig. 10. The effects of adsorbed gas molecules on flow channels and two permeability distribution in a nano-capillary (revised from Kazemi [7]).

where n_n and m_n are two dimensionless constants.

- (5) M-5: Considering the effects of formation compaction caused by increasing effective stress and matrix shrinkage due to gas desorption, Palmer and Mansoori [46] established a permeability model considering both aspects in coalbed methane reservoirs

$$k_{\sigma} = k_{ref} \left[1 + \frac{c_m}{\phi_{ref}} (p - p_{ref}) + \frac{2\varepsilon_{\ell}}{3\phi_{ref}} \frac{2v-1}{1-v} \left(\frac{\beta p}{1+\beta p} - \frac{\beta p_{ref}}{1+\beta p_{ref}} \right) \right]^3 \quad (7)$$

where v is Poisson's ratio; ε_{ℓ} and β are parameters of Langmuir curve matched to volumetric strain change due to matrix shrinkage, ε_{ℓ} is dimensionless and β is in Pa^{-1} ; c_m can be expressed by the following equation

$$c_m = \frac{(1+v)(1-2v)}{E(1-v)} - \left[\frac{2}{3} \left(\frac{2v-1}{1-v} \right) + f_p \right] c_{grain} \quad (8)$$

where E is Young's modulus, Pa; f_p is a constant between 0 and 1; c_{grain} is grain compressibility, Pa^{-1}

In Fig. 8, the commonly used stress dependent permeability models are compared. If ε_{ℓ} is designated the value of zero, then the method of M-5 only considers formation compaction due to stress dependent, and ignores the effect of matrix shrinkage caused by desorption. As we can see, all models can produce similar results to describe permeability dependent on pore pressure during the production lifetime. However, if matrix shrinkage caused by gas desorption is considered, then only the model of Palmer

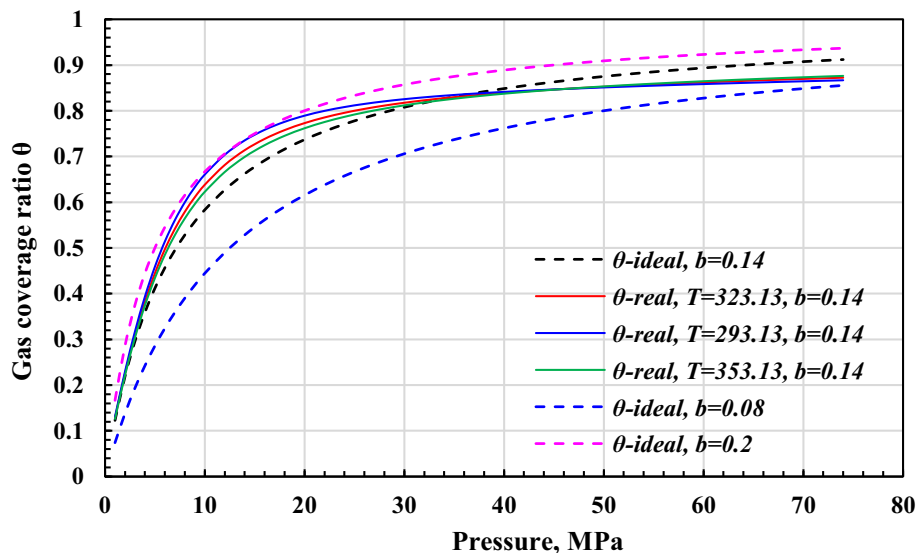


Fig. 9. Gas coverage ratio of real gas and ideal gas with different pressure, temperature and Langmuir parameter b .

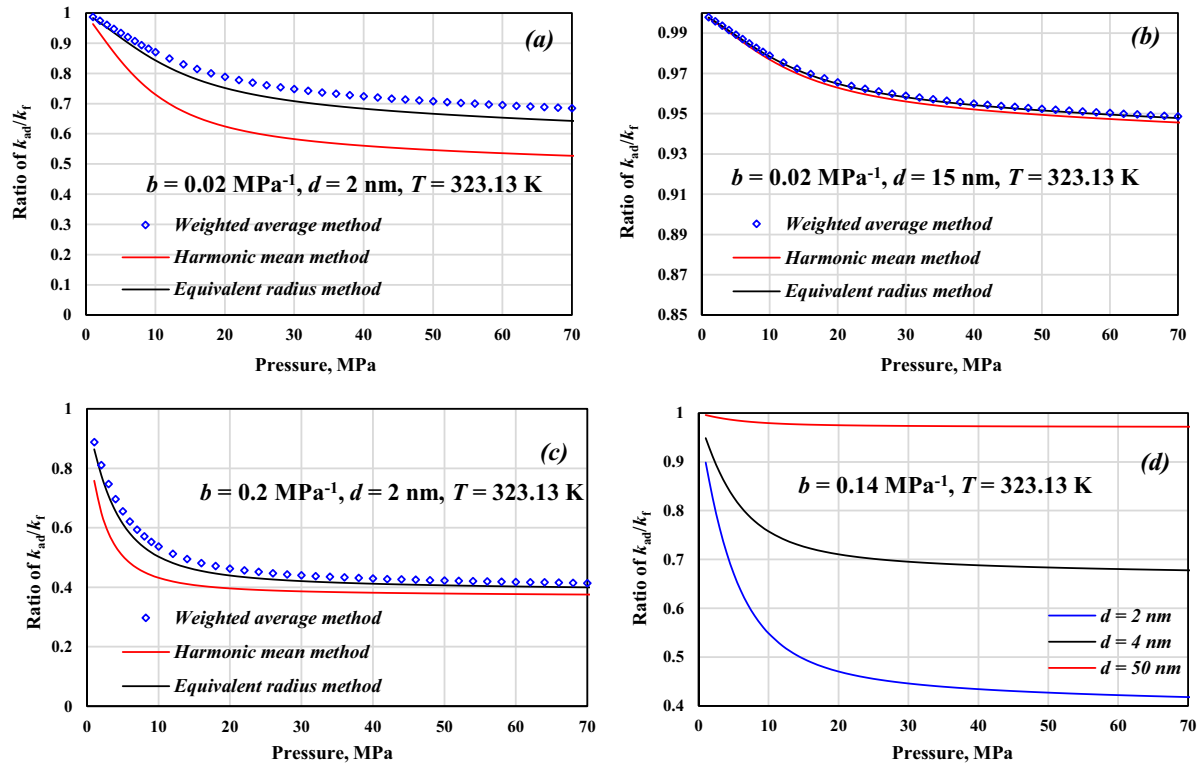


Fig. 11. The effect of adsorbed gas layer on apparent permeability: method comparison (a) and pore size effect (b).

and Mansoori [46] could be utilized to describe this phenomenon, which may have a permeability rebound at the late production period. Considering gas desorption is a common phenomenon in shale gas reservoir, the model of Palmer and Mansoori [46] is recommended to be used in practical application.

4. Effects of adsorption and desorption on apparent permeability

As mentioned, adsorbed gas is a unique gas storage form in shale formations, since organic nanopores are very developed, which provide enormous adsorption sites for gas adsorption. In nanopores, the size of gas molecular becomes comparable to the diameter of flow paths. Thus, the effect of molecular size on gas migration cannot be ignored [9]. During shale's depletion, organic pore radius can change or remain approximately constant, depending on the rates of adsorption and desorption [47]. Gas adsorption is usually assumed to satisfy Langmuir adsorption isotherm when studying the effect of gas sorption on pore radius. Langmuir adsorption isotherm for ideal gas can be expressed as

$$\theta_{ideal} = \frac{bp}{1 + bp} \quad (9)$$

while the real gas adsorption isotherm is expressed by introducing gas deviation factor Z into Eq. (9) for most practical applications [9,12,22,48,49]

$$\theta_{real} = \frac{bp/Z}{1 + bp/Z} \quad (10)$$

where θ_{ideal} and θ_{real} are gas coverage of ideal gas and real gas respectively.

Note that, Eqs. (9) and (10) are based on the assumption of ideal porous media with homogeneous surface chemistry and identical pore size, which were widely adopted in micro seepage mecha-

nism study in shale gas reservoirs [9,12,22,48–50]. However, the assumption hardly stands for strongly heterogeneous shales [43,51,52], which is characterized by complex mineral compositions (clay minerals, kerogen) [53,54] and variable flow paths with different pore scales [55] and geometries [5]. Considering the fact that Eqs. (9) and (10) based on the homogeneous assumption lead to constant adsorption heat, which is not in accordance with practical conditions, more accurate transporting models based on multi-site adsorption theory [53] are recommended to be established in future studies. Another issue needs to be noted in Eq. (10) is that the fugacity is better to be adopted for the description of chemical potential equilibrium between bulk and adsorbed phase considering real gas effects. Unfortunately, no related research, as far as we know, has been reported in shale gas seepage behavior studies currently. Thus, a more theoretically robust model based on the conception of fugacity is in need to be established for more accuracy.

Real gas effects and Langmuir parameter b influence gas coverage ratio, as shown in Fig. 9. Bigger gas Langmuir parameter b brings larger gas coverage. The effects of temperature on real gas coverage ratio are also shown in Fig. 9, which is in pressure range of 5 MPa to 30 MPa. Beyond this pressure range, the influence of temperature on real gas coverage can be ignored.

4.1. Equivalent radius method

The equivalent radius occupied by adsorbed gas molecules can be expressed as

$$r_{ad} = d_m \theta \quad (11)$$

where r_{ad} is equivalent pore radius occupied by adsorbed gas molecules; d_m is molecular diameter; θ is gas coverage for ideal gas or real gas as in Eqs. (9) and (10) respectively.

Table 1
Typical EOS of describing real gas behaviors.

Types	EOS expression	Z_c	Pons	Cons
vdW	$p = \frac{RT}{V-b_e} - \frac{a_e}{V^2}$	0.375	Simple; basis of other EOS	Low accuracy; seldom practical application
RK	$p = \frac{RT}{V-b_e} - \frac{a_e}{T^{0.5}V(V+b)}$	0.333	Practical; accurate to calculate gas phase volume	Failed to calculate liquid volume accurately
SRK	$p = \frac{RT}{V-b_e} - \frac{a_e(T)}{V(V+b_e)}$	0.333	Calculate gas and liquid phase equilibrium; widely used	The error is big when calculating liquid volume
PR	$p = \frac{RT}{V-b_e} - \frac{a_e(T)}{V(V+b_e)+b(V-b_e)}$	0.307	Higher accuracy than SRK to calculate liquid volume	Z_c is slightly bigger than practical value
Virial	$Z = \frac{pV}{RT} = 1 + \frac{B_e}{V} + \frac{C_e}{V^2}$		Able to describe viscosity, sound velocity and heat capacity of gases	Failed to calculate liquid volume; inaccurate in high pressure

Note: a_e and b_e are energy and volume parameters, respectively, and their expressions vary in different EOS; Z_c is critical gas deviation factor.

Thus, the effective pore radius (see on Fig. 10) can be expressed as

$$r_{\text{eff}} = r - r_{\text{ad}} \quad (12)$$

where r_{eff} is the effective pore radius for gas flow in shale nanopores.

Then, the effective pore radius r_{eff} will be used to establish permeability models of organic nanopores in shale gas reservoirs. The equivalent permeability considering the effect of adsorbed gas on pore occupy in a nanoscale capillary can be expressed as

$$k_{\text{ad}} = \frac{r_{\text{eff}}^2}{8} = \frac{(r - r_{\text{ad}})^2}{8} \quad (13)$$

4.2. Averaging method

The average method is to obtain the equivalent permeability considering the effect of adsorbed gas molecules on flow paths by averaging the permeability of non-adsorbed channel k_f and adsorbed channel k_a . The effect of adsorbed gas molecules on reducing the flow paths of cross section can be seen on Fig. 10.

(1) Weighted average method:

The equivalent permeability considering the adsorbed gas molecules can be expressed as a function of surface coverage θ , non-adsorbed channel permeability k_f and adsorbed channel permeability k_a , where the surface coverage θ is played as weight coefficient

$$k_{\text{ad}} = \theta k_a + (1 - \theta)k_f \quad (14)$$

(2) Harmonic mean method:

Different from weight average method of Eq. (14), Harmonic mean method was employed by Kazemi and Takbiri-Borujeni [7] to take the effect of adsorbed gas molecules on flow channels into account

$$k_{\text{ad}} = \frac{1}{\frac{\theta}{k_a} + \frac{(1-\theta)}{k_f}} \quad (15)$$

The effects of adsorbed gas molecules on apparent permeability of nanopores are shown in Fig. 11. As we can see, all three methods produce similar results in describing adsorbed gas influence on apparent permeability under large pore diameter (Fig. 11b) or large Langmuir pressure conditions (Fig. 11c), though the results from Harmonic mean method is a little smaller than those from the other two. However, the difference becomes obvious when Langmuir pressure and pore diameter are both small (Fig. 11a). Fig. 11(d) depicts the influence of adsorbed molecules on apparent permeability under different pore size conditions. Obviously, there is a significant difference between large pore and small pore cases. In pores with diameter equaling to 2 nm, the apparent permeability becomes 40–50% of non-adsorbed gas permeability k_f when pressure increased from 0 MPa to 20 ~ 40 MPa. With diameter increased to 4 nm, this ratio becomes around 70%. In macropores (larger than 50 nm), the permeability is basically not affected by gas adsorption.

5. Gas deviation factor to account for real gas effects

5.1. Real gas effects

The high pressure and temperature conditions of shale gas reservoirs underground lead to remarkable real gas effects for gas flow in in-situ formations, which changes the physical properties of gases and then changes the gas behaviors underground. In this situation, gas molecular volume and intermolecular forces cannot be ignored and a real gas transporting models need to be established to account for this behavior.

Michel et al. [56] accounted for the real gas behaviors by modifying conventional definition of the mean free path and gave a new expression to describe real gas behaviors at a large pressure range, though it was not fully validated. Wu et al. [12] and Song et al. [22] adopted this definition to calculate mean free path of real gas and established gas transporting models in shale gas reservoirs respectively. Real gas effects will also change the expression of

Table 2
Parts of methods to calculate compressibility factor Z in literatures.

Author	Method
Loebenstein [63]	$Z = 1 + \frac{p_{pr}}{10.24T_{pr}} \left[2.16 \frac{1}{T_{pr}} \left(\frac{1}{T_{pr}} + 1 \right) - 1 \right]$
Mahmoud [64]	$Z = 0.702p_{pr}^2 e^{-2.5T_{pr}} - 5.524p_{pr} e^{-2.5T_{pr}} + 0.044T_{pr}^2 - 0.164T_{pr} + 1.15$
Rayes et al. [62] method	$Z = 2.24353 - 0.0375281p_{pr} - \frac{3.56539}{T_{pr}} + 0.000829231p_{pr}^2 + \frac{1.53428}{T_{pr}^2} + 0.131987 \left(\frac{p_{pr}}{T_{pr}} \right)$
H-S-M method [61]	$Z = \frac{A_{s1} + A_{s2} \ln p_{pr} + A_{s3} (\ln p_{pr})^2 + A_{s4} (\ln p_{pr})^3 + \frac{A_{s5}}{T_{pr}} + \frac{A_{s6}}{T_{pr}^2}}{1 + A_{s7} \ln p_{pr} + A_{s8} (\ln p_{pr})^2 + \frac{A_{s9}}{T_{pr}} + \frac{A_{s10}}{T_{pr}^2}}$

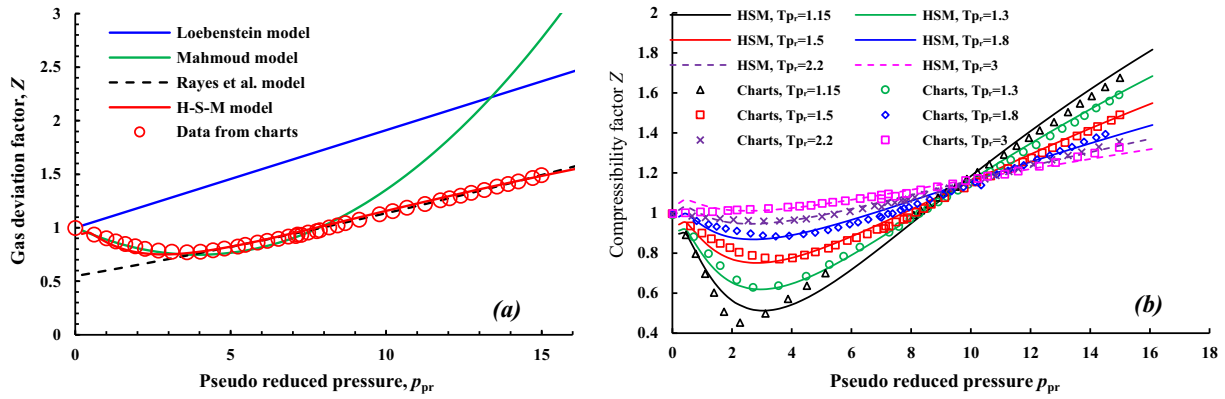


Fig. 12. The selection of gas deviation factor calculation method (a) and model further validation of the chosen model (b).

Knudsen diffusion, as will be discussed in part 8 Knudsen diffusion. Generally, the EOS of real gas will be employed, rather than EOS of ideal gas, when considering real gas effects. The adopted EOS of real gas can be semi-empirical formulae in Table 1, or empirical formula of Eq. (16), introducing a deviation factor Z to account for real gas effect. The semi-empirical formulae need to be solved by numerical techniques, while empirical formula Eq. (16) is commonly employed in the establishment of analytical flow models. Detailed calculation methods of deviation factor Z will be introduced in Part 5.2 Pseudo critical points and gas deviation factor calculation.

The empirical EOS for real gas is expressed as

$$pV = ZnRT \quad (16)$$

Besides, other parameters, such as gas viscosity, gas mean free path and Knudsen number also need to be calculated considering the effect of high pressure and high temperature conditions.

5.2. Pseudo critical points and gas deviation factor calculation

5.2.1. Pseudo critical pressure and temperature

(1) Sutton method [57]

For multi-component gases, pseudo critical properties can be calculated by the following two equations [57,58]

$$p_{pc} = (756.8 - 131.0\gamma_g - 3.6\gamma_g^2) \times 6895 \quad (17)$$

$$T_{pc} = (169.2 + 349.5\gamma_g - 74.0\gamma_g^2)/1.8 \quad (18)$$

where γ_g is specific gravity calculated by

$$\gamma_g = \frac{\sum_j y_j M_j}{M_{air}} \quad (19)$$

where y_j is gas content of the j -th gas component; M_j is molar mass of the j -th gas component; M_{air} is molar mass of air.

(2) Considering the effects of non-hydrocarbon component

In this method, the influence of non-hydrocarbon component, namely nitrogen, carbon dioxide, hydrogen sulfide on pseudo critical pressure and temperature is considered. Due to the restriction of article length, detailed procedures are ignored here and can be referred to detailed illustration in Jarrahan and Heidaryan [59] or Sutton [60], which are very simple since all of them are analytical equations. This procedure improves the accuracy of calculating pseudo critical pressure and pseudo critical temperature of multi-component gases [58], which can be calculated by the following equation respectively

$$p_{pr} = \frac{p}{p_{pc}} \quad (20)$$

$$T_{pr} = \frac{T}{T_{pc}} \quad (21)$$

5.2.2. Gas deviation factor

Gas deviation factor is one of the most important parameters in oil and gas engineering calculations, especially in those conditions with strong real gas effect, such as gas transporting behaviors in shale nanopores, which can generally be expressed as a function of pseudo reduced pressure p_{pr} and pseudo reduced temperature T_{pr} . Here, we introduce some commonly used explicit method to calculate gas deviation factor Z . Based on the digitalized Z -factor value, a correlation method was proposed by multiple regression analysis on 1220 data points in the range of $0.2 \leq p_{pr} \leq 15$ and $1.2 \leq T_{pr} \leq 3$ in the study of Heidaryan et al. [61]. A two phase compressibility factor calculation method was proposed for rich-gas-condensate system by Rayes et al. [62] based on 67 fluid-depletion studies for $0.7 \leq p_{pr} \leq 20.0$ and $1.1 \leq T_{pr} \leq 2.1$. According to statistics, there are over a hundred methods to calculate compressibility factor, some of which can be seen in Table 2.

In H-S-M method, the coefficients $A_{s1} - A_{s10}$ satisfies the following relationships, namely, $A_{s1} = 1.11532372699824$, $A_{s2} = -0.0790395208876$, $A_{s3} = 0.01588138045027$, $A_{s4} = 0.0088613449601$, $A_{s5} = -2.16190792611599$, $A_{s6} = 1.1575311867207$, $A_{s7} = -0.05367780720737$, $A_{s8} = 0.01465569989618$, $A_{s9} = -1.80997374923296$, $A_{s10} = 0.95486038773032$. A similar expression was also presented in the study of Heidaryan et al. [65] with different coefficients.

In Fig. 12(a), four methods to calculate gas deviation factor Z in Table 2 are compared to the digitalized data from classical Z -plate at the pseudo reduced temperature $T_{pr} = 1.15$. As we can see, Mahmoud model results can only match data from plate well at pseudo reduced pressure smaller than 8, while Rayes et al. model fails to match the data at pseudo reduced pressure smaller than 4. Loebenstein model results deviate from the plate data in the whole pseudo reduced pressure range. On the contrary, the matching is satisfactory between H-S-M model calculation results and Z -plate data. Thus, H-S-M method in Table 2 is recommended to calculate gas deviation factor Z in practical application. To further validate the practicability of H-S-M model, model results and data from Z -plate at different pseudo reduced temperature and pressure conditions are displayed in Fig. 12(b). The results match well on general, especially in pressure and temperature range of $1.08 < p_{pr} < 15.22$ and $1.37 < T_{pr} < 1.81$, i.e. $5 \text{ MPa} < p < 70 \text{ MPa}$ and $283.13 \text{ K} < T < 373.13 \text{ K}$.

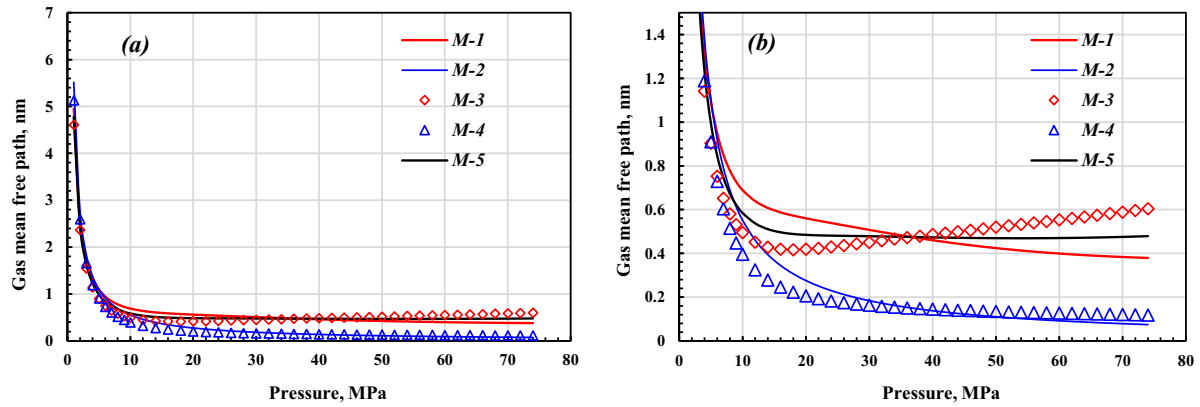


Fig. 13. Model comparison of gas mean free path calculation methods at the temperature of 323.13 K.

Table 3

Summary of different methods to calculate gas mean free path.

Method	M-1	M-2	M-3	M-4	M-5
Equation	$\lambda = \frac{\mu}{p} \sqrt{\frac{\pi RT}{2M}}$	$\lambda = \frac{k_B T}{\sqrt{2} \pi p \delta^2}$	$\lambda = \frac{\mu Z}{p} \sqrt{\frac{\pi RT}{2M}}$	$\lambda = \frac{k_B T Z}{\sqrt{2} \pi p \delta^2}$	$\lambda = \frac{\mu}{p} \sqrt{\frac{\pi Z^2 RT}{2M}}$

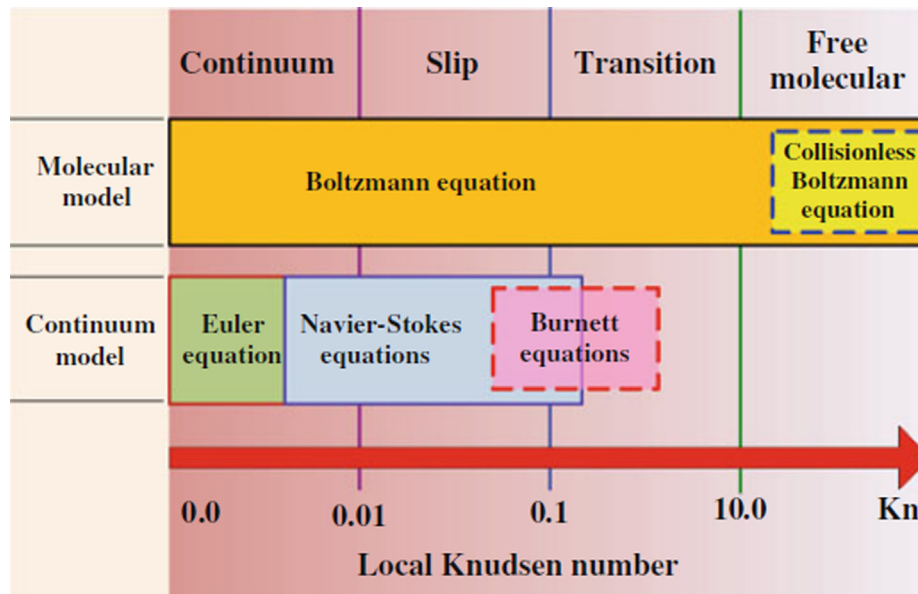


Fig. 14. Gas flow regimes at different Knudsen number conditions as well as their governing equations [84].

6. Flow chart and flow regimes division

6.1. Gas mean free path

In current study of shale gas flow behaviors, gas flow regimes are usually characterized by Knudsen number [38,66,67], which is defined as the ratio of gas mean free path to characteristic length [7,34,68,69], i.e.

$$Kn = \frac{\lambda}{L_c} \quad (22)$$

where λ is gas mean free path; L_c characteristic length, and for shale gas flow system; the pore diameter of flow paths is usually chosen as characteristic length in most previous researches [12,70] and also in this review. Pore radius r was also taken as characteristic length

in some studies [7,10,22,71–73]. This difference need to be paid attention on when applying corresponding models.

Gas mean free path λ of ideal gas can be calculated by either of the following equations [7,9,34,38,66,67]

$$\lambda = \frac{\mu}{p} \sqrt{\frac{\pi RT}{2M}} \quad (23)$$

$$\lambda = \frac{k_B T}{\sqrt{2} \pi p \delta^2} \quad (24)$$

where k_B is Boltzmann constant, 1.3805×10^{-23} J/K; δ is collision diameter of gas, which takes 0.40 nm for methane [66]. For hard sphere models, the above equations Eqs. (23) and (24) can approximately be seen as equivalent, since the viscosity of hard sphere can be calculated as [74]

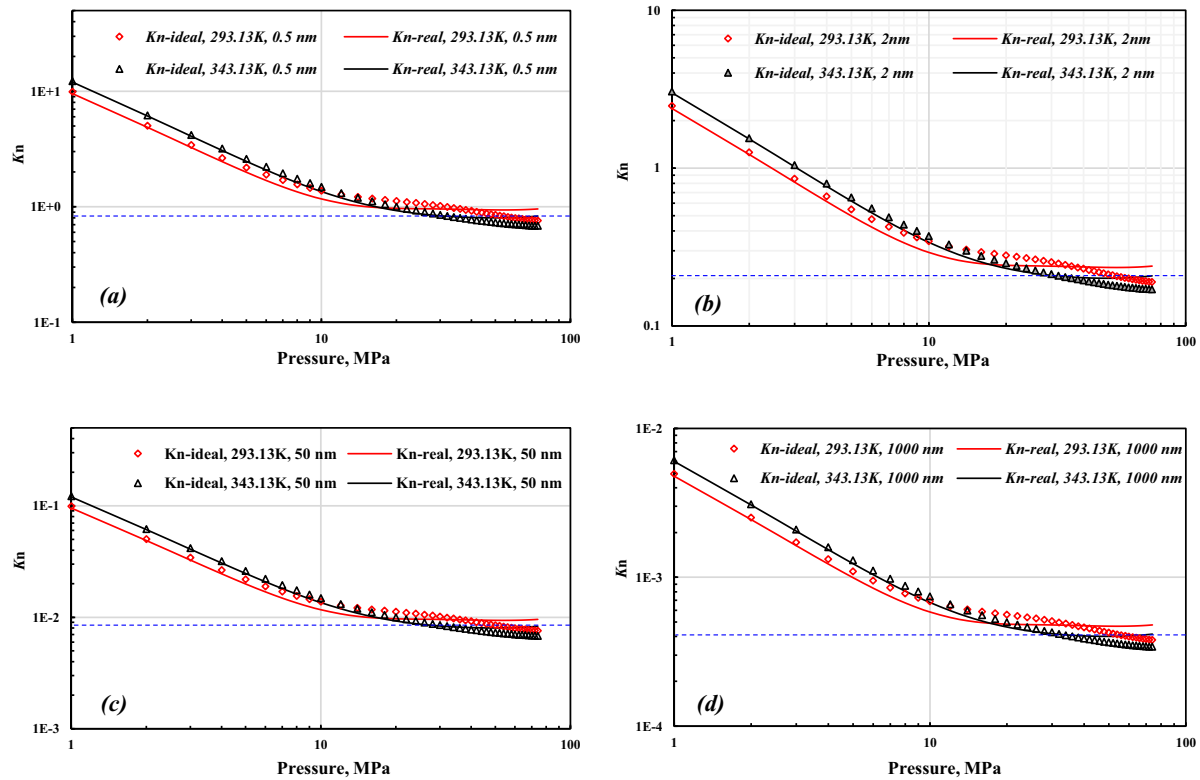


Fig. 15. Knudsen number of real gas and ideal gas in different pressure, temperature, and pore diameter conditions.

Table 4

Knudsen number range in pores with different diameters in typical pressure and temperature conditions of shale gas reservoirs.

Pore diameter	$d = 0.5$ nm	$d = 2$ nm	$d = 50$ nm	$d = 1000$ nm
Kn range	0.8–20	0.2–3	0.008–0.2	0.0003–0.006
Flow regime	Transition and Free molecular flow	Transition flow	Slip and Transition flow	Continuum and slip flow

$$\mu = 1.016 \times \frac{5}{16\delta^2} \sqrt{\frac{Mk_B T}{N_A \pi}} \quad (25)$$

Here, if we consider $16/(5 \times 1.016) \cong \pi$, and substitute Eq. (25) into Eq. (24), Eq. (23) can then be obtained. However, gas viscosity is usually obtained by empirical equations, as discussed in another review of us, in engineering application. Thus, Eqs. (23) and (24) are seen as two different methods in calculating gas mean free path.

According to Roy et al. [68], gas mean free path can be expressed as

$$\lambda = \frac{16}{5} \frac{\mu}{\rho} \frac{1}{\sqrt{2\pi RT/M}} \quad (26)$$

The EOS of real gas can also be expressed in the following form

$$\rho = \frac{pM}{ZRT} \quad (27)$$

Consider $16/5 \cong \pi$ [75] and substitute EOS of real gas of Eq. (27) into Eq. (26), the expression of gas mean free path of real gas can be expressed as [56]

$$\lambda = \frac{\mu Z}{p} \sqrt{\frac{\pi RT}{2M}} \quad (28)$$

An expression of gas mean free path for real gas was also given by Michel et al. [56], which can be seen as a modification of Eq. (24)

$$\lambda = \frac{k_B T Z}{\sqrt{2\pi p \delta^2}} \quad (29)$$

Comparing Eq. (28) to Eq. (23) or Eq. (29) to Eq. (24), a gas deviation factor Z was introduced to account for real gas effects. However, Eq. (29) can only corrects real gas behavior on molecular concentration, which ignores the correction on effective molecular diameter. It was believed to be unrelated to gas viscosity in the study of Song et al. [22]. Though molecular concentration and molecular diameter are both accounted in Eq. (28), a systematic error exists in the expression [56]. Thus, both models fail to describe real gas behaviors at high pressures. Another expression for gas mean free path was proposed [56] to overcome this issue, which was also widely adopted in corresponding studies [12,22,76]

$$\lambda = \frac{\mu}{p} \sqrt{\frac{\pi ZRT}{2M}} \quad (30)$$

Note that, the expression of gas mean free path of Eq. (14) in Civan [75] was in the following form

$$\lambda = \frac{\mu Z}{p} \sqrt{\frac{\pi RT}{2M^2}} \quad (31)$$

This is because universal gas constant R ($R = 8.314$ J/mol/K) was employed in Civan's study [75] [Eq. (13)], while it is specific gas constant R/M in Eq. (1) in the paper of Roy et al. [68], which generates the usual molecular mean free path expression of Eq. (23).

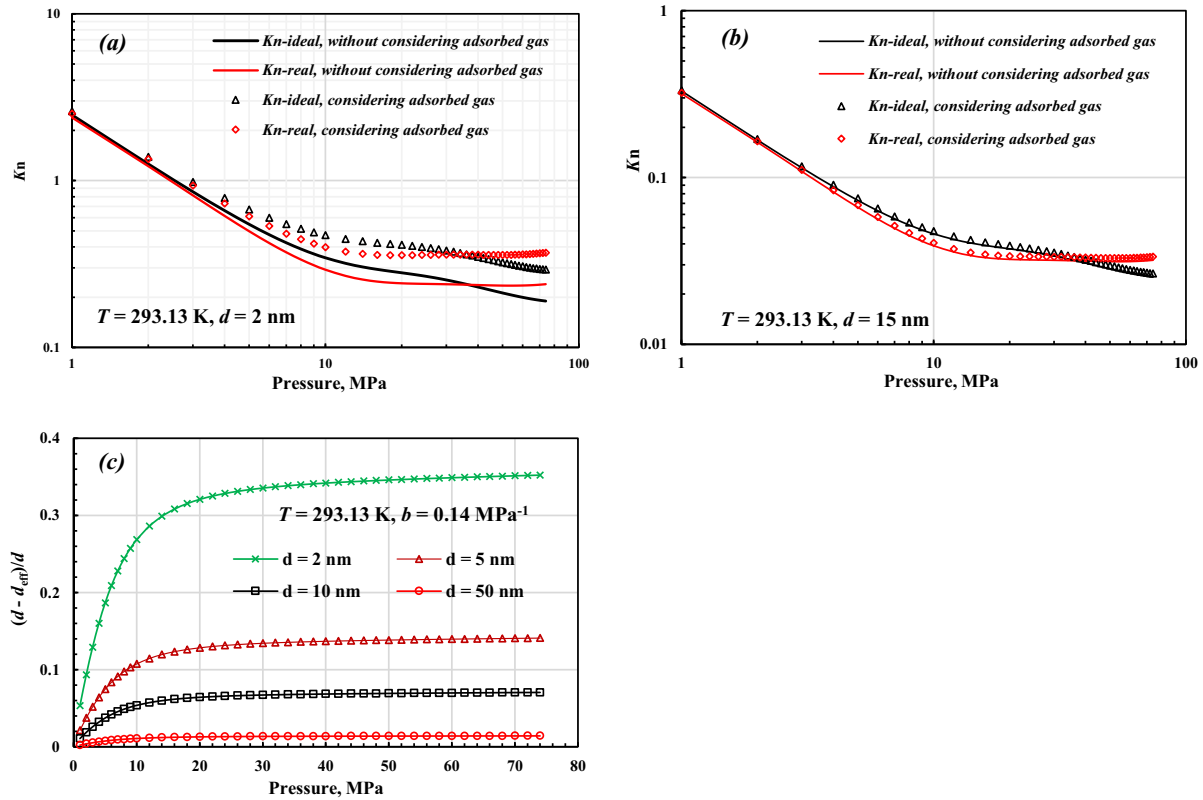


Fig. 16. Effects of adsorption on Knudsen number under different pore size conditions (note: $d_{\text{eff}} = d - 2 \cdot d_m \cdot \theta$, refer to Eq. (12), $d = 2r$ is pore diameter).

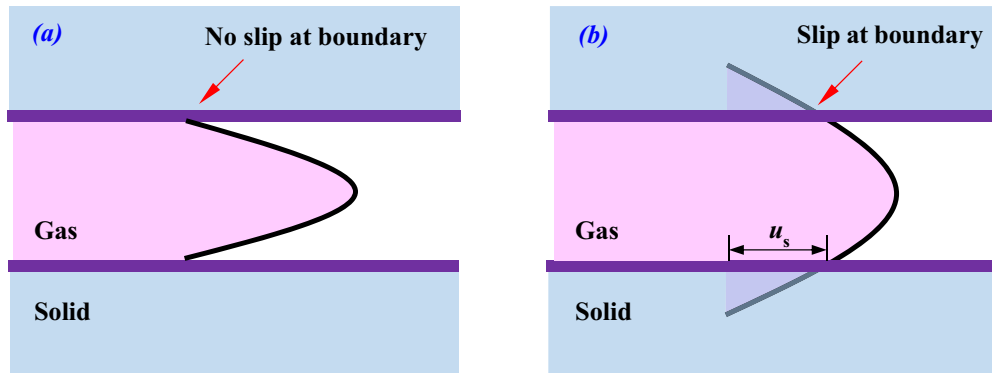


Fig. 17. Schematic diagram of gas flow without considering slippage effect (a) and considering slippage effect (b).

Different expressions of gas mean free path are compared among Eqs. (23), (24), (28), (29) and (30) and the recommended calculation method was also given after comparison.

The calculation results of gas mean free path from five different methods, as is shown in Fig. 13, are basically the same when pressure is smaller than 6 MPa, and then, the deviation occurs among different methods. The results from M -1, M -3 and M -5 are higher than those from M -2 and M -4. As reported, M -2 method in Table 3 is not very accurate at very high pressures, since the predicted gas mean free path is smaller than collision diameters at very high pressures [34]. As we can see from Fig. 13(b), Gas mean free path from M -3 increases with pressure increasing, when pressure is larger than 30 MPa, which is not physically appropriate. M -1 and M -5 methods are recommended in gas mean free path calculation for ideal gas and real gas respectively.

Table 5

The values of Klinkenberg factor b_k as a function of formation physical properties.

Reference	Expression
Heid et al. [88]	$78731.23(10^{15}k_{\text{ins}})^{-0.39}$
Jone and Owens [37]	$87139.5(10^{15}k_{\text{ins}})^{-0.33}$
Sampath and Keighin [89]	$95500(10^{15}k_{\text{ins}}/\phi)^{-0.53}$
Florence [73]	$298850(10^{15}k_{\text{ins}}/\phi)^{-0.5}$

Note: 10^{-15} is to transform the permeability unit m^2 into mD; b_k is in Pa.

6.2. Knudsen number calculation and flow regimes division

Coupling Eqs. (22), (23), (24) and (28), three kinds of Knudsen number expressions can be obtained

Table 6

The weighting coefficients of slip/viscous flow and Knudsen diffusion from different models.

Reference	ω_v for slip/viscous flow	ω_k for Knudsen diffusion	illustration
Wu et al. [12]	$1/(Kn + 1)$	$Kn/(Kn + 1)$	Double
Shi et al. [111]	$\frac{1}{1 + (Kn/Kn^{0.5})^{0.5-1}}$	$1 - \frac{1}{1 + (Kn/Kn^{0.5})^{0.5-1}}$	slippage model
Adzumi [112]	1	0.7–1.0	Single
Michel et al. [56]	1	$3\pi/16$	slippage model
Liu et al. [117]	$(1 - 2Kn)^2$	$4Kn(1 - Kn)$	
Rahmanian et al. [116]	$1 - C_A \left[1 - \exp\left(\frac{-Kn}{Kn_{lim}}\right) \right]^S$	$C_A \left[1 - \exp\left(\frac{-Kn}{Kn_{lim}}\right) \right]^S$	

Note: C_A , Kn_{lim} and S are empirical parameters, which were taken as $C_A = 2.2$, $Kn_{lim} = 0.3$ and $S = 5$ in the study.

$$Kn = \frac{\mu}{pd} \sqrt{\frac{\pi RT}{2M}} \quad (32)$$

$$Kn = \frac{k_B T}{\sqrt{2\pi p} d^2} \frac{1}{d} \quad (33)$$

$$Kn = \frac{\mu Z}{pd} \sqrt{\frac{\pi RT}{2M}} \quad (34)$$

Based on the value of Knudsen number (see Fig. 14), four flow regimes are usually characterized as continuum flow ($Kn < 0.001$), slip flow ($0.001 < Kn < 0.1$), transition flow ($0.1 < Kn < 10$) and free molecular flow ($Kn > 10$) [9,66,67]. With Knudsen number increasing, gas rarefaction effects become more and more pronounced, until the continuum assumption break down at $Kn = 0.001$. For gas flow of $Kn > 0.001$, continuum flow model needs to be modified (slip flow) or to be replaced by other governing equation or simulation method (transition flow and free molecular flow) [7,67]. Transition flow region is the most complicated and difficult flow area for simulation and modeling, which attracts much attention from scholars due to its importance. Gas flow in most shale gas reservoirs and some tight gas reservoirs fall in this region and thus, particular emphasis should be laid on the study of this subject.

Note that, the value limit of Knudsen number between viscous flow regime and slippage flow regime differs from literatures. Some studies [71,75,77–82] define viscous flow regime with Knudsen number smaller than 0.001, while others [73,83–85] define it with Knudsen number smaller than 0.01.

In Fig. 15, Knudsen numbers of real gas and ideal gas are calculated at the pressure range of 1 MPa – 74 MPa, temperature range of 293.13 K to 343.13 K. The difference of Knudsen number between real gas and ideal gas is more obvious at lower tempera-

ture and higher pressure conditions, as shown in Fig. 15. According to the values of pore diameter, Knudsen number range for real gas in pores with diameter equaling to 0.5 nm, 2 nm, 50 nm and 1000 nm is summarized in Table 4. As we can see, all flow regimes may happen in shale nanopores during production lifetime and regimes are different in pores with different pore diameters, which need to be treated respectively. Considering pores with diameters less than 2 nm is not very developed, as shown in Fig. 4, continuum flow slip flow and transition flow are main flow regimes in shale gas reservoirs.

The effects of adsorption on Knudsen number and pore diameter are shown in Fig. 16. Comparing to no adsorption cases, Knudsen number increases about 50% for pressure larger than 10 MPa in pores of 2 nm (Fig. 16a) when adsorption is considered, while it remains nearly unchanged in pores of 15 nm for whole pressure range (Fig. 16b). This is because the monolayer adsorption affects the effective pore size in small pores to more extent than that in large pores (Fig. 16c). Meanwhile, the influence is not obvious in low pressure conditions due to the small gas coverage on pore surface. Note that, the monolayer adsorption may not be valid in micro pores, since the adsorption mechanism is believed to be micro-pore filling. Corresponding studies need to be deepened to clarify the effect of adsorption on Knudsen number more accurately, since, as far as we know, current research has not distinguished the difference between micro-pore filling and monolayer adsorption on this issue, including our review.

7. Slippage effect

Darcy's law is widely employed to describe fluid flow in conventional oil or gas reservoirs, where the fluid flow is in continuum flow regime

$$J_{Darcy} = -\rho \frac{k_{ins}}{\mu} \nabla p \quad (35)$$

In tight or shale gas reservoirs, rarefaction effects become significant and the continuum media assumption becomes invalid since gas mean free path becomes comparable to flow paths. In this case, the Darcy's law of Eq. (35) needs to be modified to describe gas transfer in micropores or nanopores.

In slip flow regime, no-slip boundary condition is invalid and molecular velocity does not equal to zero, as shown in Fig. 17. This is the so called slippage effect, or Klinkenberg effect. The effect of slippage effect varies through the life of the reservoir, since the pressure changes continuously during reservoir depletion. Slippage effect may significantly affect gas flow behaviors in tight or ultra-tight gas reservoirs. Thus, various methods have been proposed to describe this phenomenon based on experimental study or theoretical study. In this part, the empirical Klinkenberg correction and slippage modification based on slip boundary conditions are introduced respectively, which are two commonly used models in considering slippage effects. For slip boundary conditions, first

Table 7

The single/double slippage models expressed in Knudsen number.

Method	Abbreviation	k_{bulk} as a function of Kn
① $F \cdot k_{ins} + k_{knudsen}$	M-1	$k_{bulk} = k_{ins} \left\{ 1 + \left[8 \left(\frac{2}{\sigma} - 1 \right) + \frac{128}{3\pi} \right] Kn \right\}$
② $\omega_v \cdot F \cdot k_{ins} + \omega_k \cdot k_{knudsen}$	M-2	$k_{bulk} = k_{ins} \left\{ \omega_v + \left[8 \left(\frac{2}{\sigma} - 1 \right) \omega_v + \frac{128}{3\pi} \omega_k \right] Kn \right\}$
③ $k_{ins} + k_{knudsen}$	M-3	$k_{bulk} = k_{ins} \left(1 + \frac{128}{3\pi} Kn \right)$
④ $\omega_v \cdot k_{ins} + \omega_k \cdot k_{knudsen}$	M-4	$k_{bulk} = k_{ins} \left(\omega_v + \frac{128}{3\pi} \omega_k Kn \right)$
⑤ $F \cdot k_{ins}$	M-5	$k_{bulk} = k_{ins} \left\{ 1 + 8 \left(\frac{2}{\sigma} - 1 \right) Kn \right\}$
⑥ $f(Kn) \cdot k_{ins} + k_{knudsen}$	M-6	$k_{bulk} = k_{ins} \left[f(Kn) + \frac{128}{3\pi} Kn \right]$
⑦ $\omega_v \cdot f(Kn) \cdot k_{ins} + \omega_k \cdot k_{knudsen}$	M-7	$k_{bulk} = k_{ins} \left[\omega_v f(Kn) + \omega_k \frac{128}{3\pi} Kn \right]$
⑧ $f(Kn) \cdot k_{ins}$	M-8	$k_{bulk} = f(Kn) k_{ins}$

Table 8
Surface diffusion coefficients from different studies (revised from Chen et al. [140]).

Reference	Pressure, MPa	Temperature, K	Surface diffusion coefficient D_s , m ² /s
Kang et al. [135]	21.14	359.00	6.35×10^{-7} m ² /s
Xiong et al. [14]	35.00	350.00	7.00×10^{-8} m ² /s
Akkutlu et al. [52]	21.14	/	4.60×10^{-7} m ² /s
Chen et al. [140]	38.00	355.15	2.56×10^{-5} m ² /s
Yang et al. [55]	6.35	355.13	2.76×10^{-20} m ² /s (average value)

order, second (or higher) order and Langmuir slip boundary conditions are classified.

7.1. Klinkenberg correction

Klinkenberg [86] correlated the permeability enhancement phenomenon of rarefied gas flow at very low pressures in tight rocks based on gas flooding experiments, where the intrinsic permeability is corrected by (detailed derivation of this equation can be referred to Florence et al. [73] or the primary source [86])

$$k_{\text{slip}} = k_{\text{ins}} \left(1 + \frac{b_k}{p} \right) \quad (36)$$

where p is average pressure across the core; b_k is slippage factor, which is usually obtained by fitting experimental data, was given by Klinkenberg [86] as a function of gas mean free path λ and pore radius r

$$b_k = \frac{4c_k \lambda p}{r} = 8c_k p Kn \quad (37)$$

where c_k is a fitting value, approximately equaling to unity [73,87].

The value slippage factor b_k in Eq. (36) can be referred to Table 5. Note that, the values of slippage factor b_k were obtained by matching with experimental data from specific reservoirs. Characteristics of tight or shale gas formations in different areas differs significantly around the world. As we can see from Fig. 18, slippage factor varies significantly from different models. For example, Florence model [73] is 3.5 – 7.5 times larger than Jones and Owens model [37] for intrinsic permeability $k_{\text{ins}} = 0.0001$ mD and porosity from 0.05 to 0.25, while slippage factor b_k from Florence model is the largest when intrinsic permeability is greater than 20 mD. There is no point in comparing these results, since they are from different reservoirs. Thus, flooding experiments are recommended to be carried out to get the value of slippage factor b_k of the specific formation before applying Klinkenberg correction of Eq. (36) on it. The slippage factor b_k increases with porosity increasing under the same intrinsic permeability condition. In addition, slippage factor b_k from all models are smaller than 0.1 MPa when intrinsic permeability is larger than 3 mD and porosity ranges from 0.05 to 0.25. This implies that slippage effect is more obvious in low permeability rocks and it can be ignored when intrinsic permeability is larger than 3 mD. Meanwhile, the expressions of slippage factor b_k in Table 5 are only functions of solid properties and are independent of pressure, thus, real gas considerations do not affect slippage phenomenon and the value of slippage permeability k_{slip} .

Note that, the square-root correlation of Florence model [73] is developed from a theoretical basis, rather than regression from experimental dataset, which is based on KoZeny-Carman formula of the relationship among permeability, porosity and equivalent pore radius and slippage factor b_k from Klinkenberg's study, namely, Eq. (37). Thus, it is supposed to have a broader application scope than the other three methods [37,88,89]. However, a constant temperature of 298 K, ambient pressure 0.1 MPa and Nitrogen gas properties (molar mass, viscosity, gas mean free path) were used during the deriving process. Thus, for high temperature gas slippage study, a new expression of slippage factor b_k is recom-

Table 9
Different values of first order slip velocity coefficient C_1 in Eq. (41).

Reference	First order slip coefficient C_1	Approach	Kn range
Maxwell [94]	1	MD theory	$Kn \leq 0.1$
Ohwada et al. [141]	1.114	LBE for hard sphere molecules and diffuse BC, solved by finite difference method	$Kn \leq 0.1$
Marques [142]	1.111	Theoretical analysis of five field equations on steady plane Couette flow	$Kn \leq 0.25$
Albertoni et al. [121]	1.1466(5)	Numerical quadrature using analytical formula obtained from BGK kinetic model	$Kn < 0.1$
Welander [121]	1.23	Numerical solution of an integral equation	/
Willis [121]	1.142	Exact formula from the previous equation	/
Cercignani [121]	1.136	Expansion in eigen-functions	/
Wang Chang, Uhlenbeck [121]	1.134	Expansion in the semirange of Maxwell molecules	/
Gross [121]	1.152	Expansion in the semirange of BGK model	/
Ziering [121]	1.181	Expansion in the semirange	/
Siewert [84,143]	1.1478	Polynomial expansion and analytical discrete-ordinates method for Cercignani-Lampis boundary condition	$Kn < 0.1$
Wakabayashi et al. [84]	1.1114	LBE	$Kn < 0.1$
Loyalka [119]	$0.5 \pi^{0.5} (1 - 0.1817\sigma)$	BGK kinetic model	$Kn < 0.1$
Loyalka et al. [120]	$0.5 \pi^{0.5} (1 + 0.1621\sigma)$	BGK model and Maxwell diffuse specular reflection	$Kn < 0.1$
Lockerby et al. [95]	$(2/\pi)^{0.5}$	Wall function method with Knudsen layer structure captured for locally non-equilibrium gas flows	$Kn < 0.1$
Lilley and Sader [144]	$\frac{1}{2Kn} \left(\frac{1}{\gamma_L} - 1 \right)$	Using LBE solutions and DSMC calculations, where γ_L is normalized mid-plane velocity gradient	$Kn < 0.1$
Fichman and Hetsroni [78]	$1 + 3 Kn + 2 Kn^2/3$	Considering gas viscosity reduction in Knudsen layer on Maxwell BC	$Kn < 0.3$
Bahukudumbi et al. [122]	$1.2977 + 0.71851 \arctan(-1.17488 Kn^{0.58642})$	Empirical model based on linearized Boltzmann solutions and DSMC results	$Kn \leq 12$
Karniadakis et al. [71,145]	$1/(1 - b Kn)$	N-S and DSMC method	Entire
Agrawal and Prabhu [77]	1.13	Analysis on experimental data of cylindrical Couette flow	(0.04, 0.1)
Pan et al. [81]	1.70 1.1254	Qualitative analysis of Couette flow using DSMC simulations	(0.1, 8.3) $Kn < 0.1$

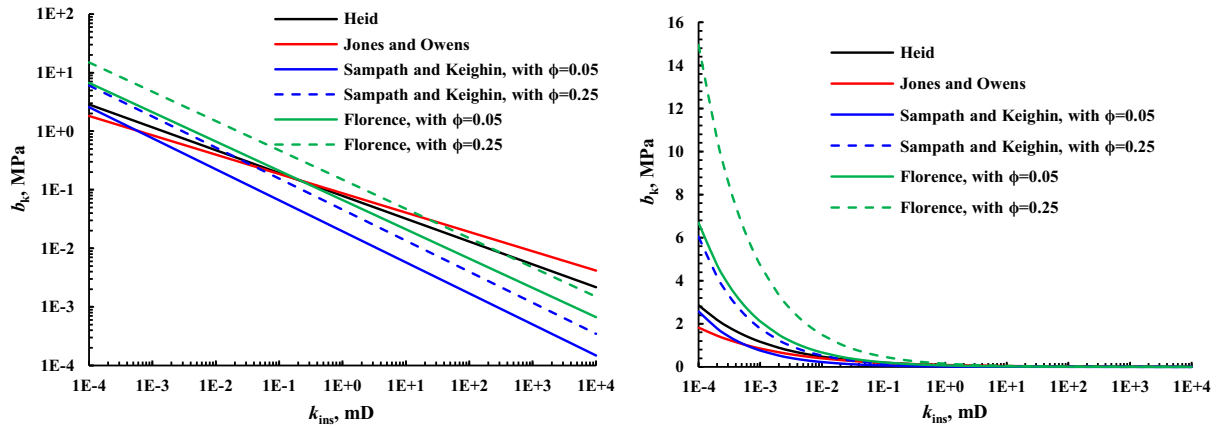


Fig. 18. Slippage factor b_k variance with intrinsic permeability in different models.

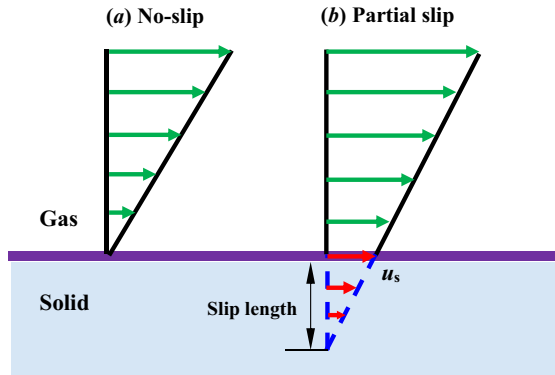


Fig. 19. Comparison of no-slip and slip boundary conditions (revised from Zhang et al. [84]).

mended to be derived according to the method of Florence et al. [73] for the specific gas type (mainly methane for shale gas reservoirs). Besides, the parameter tortuosity in KoZeny-Carman formula was ignored in the square-root correlation of Florence model [73].

Compared with slip boundary conditions discussed later, Klinkenberg correction of Eq. (36) is equivalently first order [66,73], since Knudsen number Kn is proportional to reciprocal of pressure $1/p$. It was also derived from first order approximation mathematically [73]. Obviously, the slippage phenomenon is more remarkable at low pressure conditions, generating a larger value of k_{slip} . Klinkenberg correction method is based on Warburg's model, which assumes the mean free path is small compared with capillary radius [37]. The plot of slippage permeability versus reciprocal pressure might not display a straight line for low permeability tight gas or shale gas reservoirs due to the depressions in slippage factor b_k determined at reduced pressure departing from Warburg's model [86]. Based on experimental data of more than 100 tight gas sand samples, an exponential relationship between slippage factor b_k and intrinsic permeability k_{ins} was found, as shown in Table 5.

For simplification, a correction factor F was introduced to modify the intrinsic permeability k_{ins} in the following form

$$k_{slip} = Fk_{ins} \quad (38)$$

correspondingly, the mass flux considering Klinkenberg modification can be expressed as

$$J_{slip} = FJ_{Darcy} \quad (39)$$

Brown et al. [90] studied gas flow in pipes at low pressures and proposed a theoretical form of correction factor as

$$F = 1 + \frac{\mu}{pr} \left(\frac{2}{\sigma} - 1 \right) \sqrt{\frac{8\pi RT}{M}} \quad (40)$$

where p is average pressure; σ is tangential momentum accommodation coefficient (TMAC), which denotes the percentage of gas molecules reflected diffusively from the wall, ranging from 0 to 1. Experimental study or molecular dynamics need to be done for the determination of tangential momentum accommodation coefficient σ , which is influenced by wall surface smoothness, gas type, temperature and pressure [18].

Another issue needs to be noted is that the initial water saturation always [44,91] exists in shale gas reservoirs, which affects both adsorption capacity and gas flow behaviors [43,49,92]. It has been experimentally investigated that gas slippage factor increases with water saturation increasing [93]. Meanwhile, the heterogeneous pore structures may lead to complex water distribution characteristics underground, which makes the problem even harder to be dealt with. Considering the effects of irreducible water, Wang et al. [44] pointed out apparent permeability decreases with the increase of relative humidity, and water blockage would lead to a sharp decrease of apparent permeability in high humidity conditions. Thus, slippage effect needs to be systematically studied coupling the effects of pore structures, surface heterogeneity, gas–water two phase flow conditions *et al.* in the future.

7.2. Slip boundary conditions

Slip boundary conditions describe the velocity slip at the wall surface for gas flow with Knudsen number $Kn > 0.001$, where the no-slip boundary condition is no longer valid, as can be seen from Fig. 19. Slip boundary conditions are expressed as a function of Knudsen number Kn , which can not only capture velocity slip at pore surface during slip flow regime, but characterize rarefaction effects for gas flow in transition flow regime, or even free molecular regime. Thus, slip boundary conditions are of great significance for gas flows with large Knudsen number.

7.2.1. First order slip boundary condition

Maxwell [94] proposed a first order slip boundary condition for gas flow in nanoscale or microscale flow channels as

Table 10Different values for first and second order slip coefficient C_1 and C_2 in Eq. (44).

References	C_1	C_2	Approach/Annotation
Schamberg [84,128]	1	$5\pi/12$	Second order model
Karniadakis and Beskok [84,125]	1	−0.5	Second order model
Ng et al. [126]	1.15	$0.25 \text{Kn}^{-0.65}$	Multi-coefficient slip model using Taguchi quality control techniques and numerical analysis, suitable to transition regime
Cercignani [84,146]	1.1466	0.9756/0.647	Linearized Boltzmann-BGK model
Chapman and Cowling [84,147]	1	0.5	Linearized Boltzmann model
Hadjiconstantinou [148]	1.11	0.61	Hard sphere model
	1.11	0.31	Hard sphere model with Knudsen layer
Deissler [96]	1	9/8	Second order model from physical derivation
Cercignani [84,149]	0.8297	0.5108	Boltzmann-BGK model
Hsia and Domoto [131]	1	0.5	Second order model of modified Reynolds equation based on experimental investigation
Mitsuya [150]	1	2/9	1.5 order modified Reynolds equation, TMAC has no effect on second order slip flow term
Lockerby et al. [124]	1	0.145–0.19 (depend on Pr)	Maxwell–Burnett model for Poiseuille flow
Wu and Boggy [151]	1	1	Modified Reynolds equation for compressible and extremely thin layer gas lubrication [151]
Wu and Boggy [129]	2/3	1/4	New second order model
Sun et al. [152]	0.62228	0.1936	Variable hard sphere model
	0.63875	0.204	Variable soft sphere model
Yudistiawan et al. [153]	1	2/3	D2Q9 LB model
Loyalka and Hickey [84,133,154,155]	1.1019	0.449	BGK based kinetic model
Lockerby and Reese [156]	0.798	0.278	Combining new slip boundary condition with near wall scaling of N-S constitutive relation
Kim et al. [127,157]	$(6/\pi)^{0.5}$	4/π	D2Q9 LB model for Poiseuille flow
Kim and Pitsch [157]	1.073	0.514	Higher order lattice Boltzmann method based on the forth order Gauss–Hermite quadrature
Cercignani and Lorenzani [158]	1.1209	0.2347	Variational solution of the linearized Boltzmann equation for hard sphere molecules
Struchtrup and Torrilhon [84,159]	1	0.531	R13 based model
	1.0462	0.5485	R13 based model with Knudsen layer effects
Li et al. [84,134]	1	0.8	LB model
Gibeli [133]	1.1144	0.49522	Hard sphere model with Knudsen layer effect
Kezami et al. [7]	0.2183 + 0.9233/σ	0.2825	Analytical model based on kinetic theory and using regularized 13 moment method

Table 11The slip velocity coefficients C_1 and C_2 including TMAC in Eq. (44).

References	C_1	C_2	Approach/Annotation
Beskok and Karniadakis [71]	$\frac{2-\sigma}{\sigma}$	$-\frac{1}{2}(\frac{2-\sigma}{\sigma})$	Physics based models claim to be applicable in entire Kn range
Aubert and Colin [130]	$\frac{2-\sigma}{\sigma}$	9/4	Analytical solution of N-S equations for steady gaseous flow in rectangular microducts
Karniadakis and Beskok [84,125]	$\frac{2-\sigma}{\sigma}$	$-\frac{2-\sigma}{\sigma}$	N-S equation
Hsia and Domoto [131]	$\frac{2-\sigma}{\sigma}$	$\frac{1}{2}(\frac{2-\sigma}{\sigma})$	Second order model of modified Reynolds equation based on experimental investigation
Jie et al. [79,84]	$\frac{2-\sigma}{\sigma}$	$-(\frac{2-\sigma}{\sigma})\frac{Re}{Kn}$	Solve N-S equation with slip BC using LU-TVD implicit algorithm
Roohi and Darbandi [160]	$\frac{2-\sigma}{\sigma}$	$-\frac{2-\sigma}{\sigma}$	Extending N-S equation using IP scheme accurate for Kn < 0.5
Wu and Boggy [129]	$\frac{2}{3}(\frac{2-\sigma}{\sigma})$	1/4	Second order slip model using a physical approach
Gaur et al. [84,161]	$1.012 \frac{2-\sigma}{k_s}$	$\frac{2}{k_s^2}$	Analytical solution from quasi gas-dynamic equations equation, k_s depend on molecular interaction model
Deissler [96]	$\frac{2-\sigma}{\sigma}$	9/8	Second order model from physical derivation
Cercignani [84,162]	$\frac{2-\sigma}{\sigma}(1 + 0.1621\sigma)$	$\frac{2}{\pi}(\frac{1}{2} + c_c^2)$	Boltzmann equation, c_c is a parameter
Zhang et al. [163]	$\frac{1-(1-C_2)\sigma}{\sigma}$	$-\frac{(1-C_2)}{2} \left[\frac{1-(1-C_2)\sigma}{\sigma} \right]$	Corrected second order slip boundary condition considering relative position of slip surface in Knudsen layer, $c_2 \in [0, 1]$
Wu [69]	$\frac{2}{3} \left[\frac{3-\sigma f_w^3}{\sigma} - \frac{3(1-f_w^2)}{2Kn} \right]$	$\frac{1}{4} \left[\frac{f_w^4}{Kn^2} + \frac{2}{Kn^2}(1-f_w^2) \right]$	Slip model derived from kinetic theory for rarefied gas flows in the whole Knudsen number range, $f_w = \min [1/Kn, 1]$

$$u_s - u_w = \pm C_1 \frac{2-\sigma}{\sigma} \lambda \left(\frac{\partial u}{\partial n} \right)_s \quad (41)$$

where C_1 is first order slip coefficient, which equals to unity derived by Maxwell; u_s is gas velocity on the wall; $(\partial u / \partial n)_s$ is velocity gradient along the surface normal.

The Maxwell first order slip model [94] with $C_1 = 1$ overestimates the amount of microscopic actual slip velocity, as indicated by kinetic theory and MD simulation [95]. Thus, various slip velocity coefficients have been proposed by BE, DSMC, MD simulations

or experimental measurements, as an extension of Maxwell first order slip boundary condition.

7.2.2. Second or higher order slip boundary condition

The first order slip boundary condition of Eq. (41) is relatively accurate for Knudsen number $Kn < 0.1$, while it shows considerable discrepancies against experimental data when Knudsen number $Kn > 0.1$ [83]. Second order slip boundary conditions have been proposed to improve the simulation accuracy and extend the validity of the slip flow regime to higher Knudsen numbers. Up to now, no

Table 12Different values of C_1 and C_2 in $f(Kn)$ from experimental measurements [84].

Reference	Gas type	C_1	C_2	Knudsen number range
Maurer model [100]	He	1.2	0.23	$0.06 < Kn < 0.8$
	N ₂	1.3	0.26	$0.002 < Kn < 0.6$
Sreekanth [97]	N ₂	1.0	0	$0.007 < Kn < 0.03$
		1.1466	0	$0.03 < Kn < 0.13$
		1.1466	0.14	$0.13 < Kn < 0.237$
Porodnov et al. [132]	He	1.099 ± 0.02	0	$0.03 < Kn < 0.3$
Colin et al. [164]	He	1.02	0	$0.03 < Kn < 0.25$
Ewart et al. [98]	N ₂	1.066 ± 0.088	0.231 ± 0.057	$0.003 < Kn < 0.291$
	He	1.052 ± 0.02	0.148 ± 0.014	$0.009 < Kn < 0.309$
Ewart et al. [99]	Ar	1.147 ± 0.042	0.294 ± 0.029	$0.003 < Kn < 0.302$
	He	1.26 ± 0.022	0.17 ± 0.02	$0.03 < Kn < 0.3$
Yamaguchi et al. [101]	Ar	1.3 ± 0.09	0.063 ± 0.018	$0.05 < Kn < 0.3$
	N ₂	1.35 ± 0.06	0.031 ± 0.005	/
	O ₂	0.35 ± 0.06	0.028 ± 0.004	/
				/

consensus has been reached on the expression form of second order formulation. A second order slip boundary condition was proposed as [96]

$$u_s - u_w = \pm \frac{2 - \sigma}{\sigma} \lambda \left(\frac{\partial u}{\partial n} \right)_s - \frac{9}{8} \lambda^2 \left(\frac{\partial^2 u}{\partial n^2} \right)_s \quad (42)$$

A more commonly used second order boundary condition is also written in the following form

$$u_s - u_w = \frac{2 - \sigma}{\sigma} \left[\pm \lambda \left(\frac{\partial u}{\partial n} \right)_s - \frac{\lambda^2}{2} \left(\frac{\partial^2 u}{\partial n^2} \right)_s \right] \quad (43)$$

As we can see, the tangential momentum accommodation term $(2 - \sigma)/\sigma$ affects only the first order slip coefficient in Eq. (42), while it affects both the first and the second order slip velocity coefficient in Eq. (43). On this issue, there is no conclusion on which opinion is right. From this reason, a simpler but still efficient second order boundary condition was usually adopted for fully diffusive boundaries

$$u_s - u_w = \pm C_1 \lambda \left(\frac{\partial u}{\partial n} \right)_s - C_2 \lambda^2 \left(\frac{\partial^2 u}{\partial n^2} \right)_s \quad (44)$$

where C_1 and C_2 are first order and second order slip coefficient respectively; the “+” denotes the boundary under the gas, while “-” denotes for the boundary above the gas.

Still, there is no general agreement on the values of the first and second order slip velocity coefficients, which has drawn numerous studies up to data, as can be seen in Table 10

The accuracy of second order boundary model depend on the determination of slip coefficients C_1 and C_2 . However, as we can see from Table 10, no universal slip velocity coefficients have been reached for slip boundary conditions, and some of them even have opposite tendency. The second order slip velocity coefficients in Beskok and Karniadakis and Lockerby and Reese generate a decrease in the slip velocity at the wall, while most of models generate an increase in slip velocity at the boundary. It was found that the first order slip models underestimate the mass flow rate by comparing with experimental observations, implying that the second order modification plays an enhancement on velocity at the boundary.

If the molecule reflections at the boundary are not fully diffusive, the slip velocity coefficients C_1 and C_2 are related to tangential momentum accommodation coefficient, as derived by Navier-Stokes equation, quasi-gas dynamic and Boltzmann equations. The slip velocity coefficients C_1 and C_2 as a function of TMAC are summarized in Table 11.

Besides theoretical derivation and numerical simulation, the first and second order slip coefficients C_1 and C_2 can also be measured by experiment, as reported by Sreekanth [97], Ewart et al.

[98,99], Maurer et al. [100], Yamaguchi et al. [101] and summarized by Agrawal and Prabhu [102] and Zhang et al. [84]. The results are listed in Table 12.

7.2.3. Langmuir slip boundary condition

As introduced above, gas adsorption and desorption is a unique characteristic of shale gas in underground formation owing to the interaction force between gas molecules and solid surface [84], which can be described by Langmuir adsorption theory. Based on Langmuir adsorption theory, a robust Langmuir slip boundary condition was proposed [103] taking the interfacial interaction between gas molecules and solid surface into account, which is expressed as

$$u_s = \theta u_w + (1 - \theta) u_{ref} \quad (45)$$

where u_g is reference gas velocity, such as a mean free path away from the wall or gas velocity at the mid-stream [104,105]; θ is gas coverage ratio of ideal gas or real gas as expressed in Eqs. (9) and (10), respectively.

Langmuir slip boundary condition is derived from a more physical approach, which gets rid of empirical parameter TMAC and the velocity gradient in the expression [106], as is shown in Eq. (45). Thus, it can be implemented after sorption experiment and produces a bounded velocity value. Using Langmuir slip boundary condition, Myong [105] studied the problems of pressure driven microchannel gas flow and low Reynolds number gas flow past a sphere and proved the application value of Langmuir slip model. Velocity slip on solid surfaces of microscale cylindrical Couette flow was investigated by Myong et al. [104] using Langmuir slip model for solid gas molecular interaction, where the velocity inversion was explained by introducing a velocity polar on the hodo-

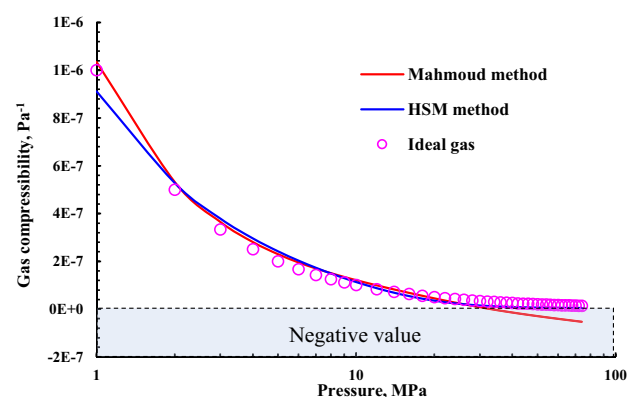


Fig. 20. Gas compressibility in different pressure conditions at the temperature of 313.13 K.

graph plane. The difference between Maxwell slip model and Langmuir slip model was also reported by Kim et al. [107] and Chen and Tian [108] using lattice Boltzmann method.

Considering the fact that gas molecules on solid surface do not rebound instantly or elastically, but interact with solid surface due to intermolecular force field, Singh and Javadpour [109] presented a single analytical permeability model to account for gas adsorption on solid surface and the consequently desorption after some time lag, which also included the effect of pore enlargement due to pressure depletion sorption. Zhang et al. [16] developed a gas transporting model in organic shale nanopores, where the slippage effect was taken into account by Langmuir slip boundary condition.

8. Knudsen diffusion

As been introduced in above section, Knudsen diffusion is usually adopted to describe fluid flows with Knudsen number $Kn > 10$ (free molecular flow) [12,110,111]. Meanwhile, the occurring of slippage effect is attributed to Knudsen diffusion in some other studies [73,87]. Knudsen diffusion is a diffusive transport process proportional to the pressure gradient, but independent of gas density [110]. In this section, the derivation of Knudsen diffusion equation as well as the equivalent permeability will be introduced, while the application of Knudsen diffusion and its role in bulk gas transport models will be discussed in the next section.

- (1) *Ideal gas*: Based on the linear relationship between flux and pressure drop, a diffusive transport equation can be obtained with a constant diffusion coefficient and negligible and viscous effects, which is [68,112]

$$J_{Knudsen} = \frac{q}{A} \rho = -\frac{MD_k}{RT} \nabla p \quad (46)$$

where $J_{Knudsen}$ is mass flux of Knudsen diffusion, $\text{kg}/(\text{m}^2 \cdot \text{s})$; q is volume flow rate, m^3/s ; D_k is Knudsen diffusion coefficient, m^2/s , which is expressed as

$$D_k = \frac{d}{3} \sqrt{\frac{8RT}{\pi M}} \quad (47)$$

Javadpour et al. [34] noted that the continuum flow ($Kn < 0.001$) can be described by Darcy's law, while it is slip flow for $0.001 < Kn < 0.1$ and needs to be described by Knudsen diffusion. In this paper [34], the Knudsen diffusion of Eq. (46) was introduced to describe gas flow in nanopores, and the superposition of viscous flow considering slippage effect and Knudsen diffusion was proposed for the first time in shale gas reservoir study. However, this is not the first to consider dual-mechanisms, namely laminar process and Knudsen flow process, of gas in very tight porous media. Adzumi [112] studied flow behaviors of gaseous mixtures through capillaries with transitory pressure range by adding the Hagen-Poiseuille flux and Knudsen diffusion flux, where the Knudsen diffusion term was multiplied by a constant value proportional to the relative volume of the mobile boundary layer. Later, the method was extended into modeling slippage effect by Ertekin et al. [87] in tight porous media by linear superposition of viscous flow and Knudsen diffusion, and proposed a dual-mechanism dynamic gas slippage factor. Note that, the molecular mass M in the first term of Eq. (3) in the paper [34] should be at the denominator, rather than numerator. Since then, Knudsen diffusion has been widely employed in the study of shale gas migration mechanisms.

Comparing Eq. (46) to Eq. (35) and using EOS of ideal gas, the equivalent permeability of Knudsen diffusion can be expressed as

$$k_{Knudsen} = \frac{\mu D_k}{p} \quad (48)$$

- (2) *Real gas*: Knudsen diffusion can be modelled by Fick's law of diffusion when gas is assumed to be traveling under concentration field. For single component gas, self diffusion is supposed to be happening under the influence of concentration gradient or, in other words, density gradient, which are the same for single phase flow.

$$J_{knudsen} = -MD_k \nabla C \quad (49)$$

where c is gas molar concentration, mol/m^3 , which have the following relationship according to EOS of real gas

$$C = \frac{p}{ZRT} \quad (50)$$

where only p and Z varies with location. Thus, substituting Eq. (50) into Eq. (49), we can obtain

$$\begin{aligned} J_{knudsen} &= -\frac{MD_k}{RT} \nabla \left(\frac{p}{Z} \right) = -\frac{MD_k}{RT} \frac{p}{Z} \left(\frac{1}{p} - \frac{1}{Z} \frac{dZ}{dp} \right) \nabla p \\ &= -\frac{MD_k}{RT} \frac{p}{Z} c_g \nabla p \end{aligned} \quad (51)$$

Comparing Eq. (51) with Darcy's law, the equivalent permeability of Knudsen diffusion for real gas can be written as

$$k_{knudsen} = D_k c_g \mu \quad (52)$$

For ideal gas, $c_g = 1/p$, the equivalent permeability of Knudsen diffusion for real gas in Eq. (52) reduced to ideal gas permeability in Eq. (48).

For real gas compressibility, it can be calculated by the correlation developed by Mahmoud [64]

$$\begin{aligned} c_g &= \frac{c_{pr}}{p_{pc}} \\ &= \frac{1}{p_{pc}} \left[\frac{1}{p_{pr}} - \frac{1}{Z} (1.404 p_{pr} e^{-2.5 T_{pr}} - 5.524 e^{-2.5 T_{pr}}) \right] \end{aligned} \quad (53)$$

However, the Mahmoud method Eq. (53) to calculate gas compressibility is based on the calculation method of gas deviation factor Z in Table 2, which was proved inaccurate in high pressure conditions. Thus, the H-S-M method in Table 2 was adopted to calculate gas compressibility in this paper. As we can see from Fig. 20, gas compressibility becomes negative at high pressure conditions ($p > 40$ MPa), which is physically inappropriate. Thus, H-S-M method is still recommended to calculate gas compressibility.

- (3) Another way to study Knudsen diffusion in literatures is through the difference between the bulk phase gas transporting permeability k_{bulk} and intrinsic permeability, which can mathematically be expressed as

$$k_{knudsen} = [f(Kn) - 1] k_{ins} \quad (54)$$

where $k_{bulk} = f(Kn) \cdot k_{ins}$ represents the superposition effects of viscous flow and Knudsen diffusion, and some studies also consider that this part belongs to the contribution of slip flow, as will be discussed later. In essence, this study method is a type of single slippage model in Part 3.5.4.2.

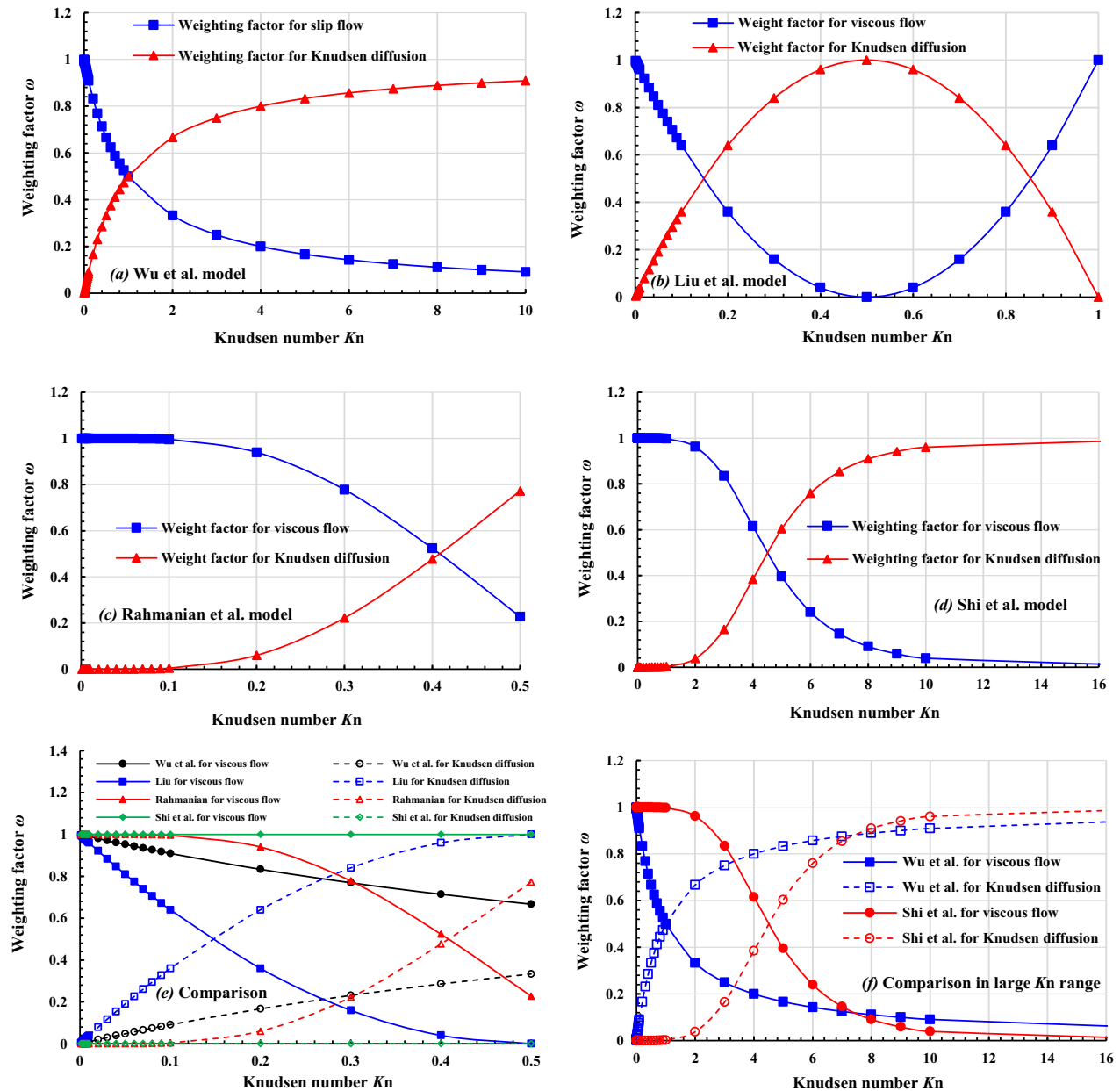


Fig. 21. The variance of weighting factor with Knudsen number in different models.

The role of Knudsen diffusion in shale gas migration study is still unclear and very confusing, which needs to be investigated in depth in the future.

9. Bulk phase gas transport models

In this part, two kinds of bulk phase gas transport models are introduced. One is based on the superposition of slippage flow and Knudsen diffusion, while the other can be derived from slip boundary conditions and be expressed as a function of Knudsen number.

9.1. Superposition models

9.1.1. Double slippage model

Double slippage model [111] in this paper refers to that the model considers Klinkenberg effect and Knudsen diffusion simulta-

neously, i.e. both Klinkenberg correction and Knudsen diffusion are supposed to be the reason of slippage effect [12,47]. In some studies, Knudsen diffusion is employed to describe free molecular regime, while slippage effect is also considered [113]. This kind of models are also classified to double slippage model here, since adopting Knudsen diffusion equation to describe slippage effect and to describe free molecular flow are the same in essence from a mathematical point of view. According to the difference of superposition modes, double slippage model can be classified into linear superposition model and weight superposition model.

(1) Linear superposition [10,22]

Linear superposition of viscous/slippage flow and Knudsen diffusion can be seen as a kind of dust gas model, with weighting coefficient for either component equaling to unity, as in Eqs. (55), (56) and (62). As will be derived and discussed later, the linear superposition is also the first order approximation for gas flow in nanopores.

Javadpour developed an analytical model considering diffusive and advective gas flow in shale rocks, where the total gas flux was determined by linear superposition of Knudsen diffusion and viscous flow with Klinkenberg effect [18,114].

$$J_{\text{Bulk}} = FJ_{\text{Darcy}} + J_{\text{Knudsen}} = J_{\text{slip}} + J_{\text{Knudsen}} \quad (55)$$

J_{Bulk} is mass flux of bulk phase gas transport in nanopores.

Substituting Eqs. (39) and (46) into Eq. (55) and comparing it to Darcy's law of Eq. (35), the equivalent permeability of bulk phase transport in nanopores can be expressed as

$$k_{\text{bulk}} = Fk_{\text{ins}} + k_{\text{Knudsen}} \quad (56)$$

According to different expressions of correction factor F and Knudsen equivalent permeability k_{Knudsen} , different forms of equivalent bulk phase permeability k_{bulk} can be obtained.

The Knudsen diffusion was introduced into modeling shale gas transport in nanopores from the study of Roy et al. [68] for the first time by Javadpour et al. [11]. Later, Javadpour [18] combined the slippage viscous flow and Knudsen diffusion as in Eq. (55) to describe different gas migration mechanisms in nanopores or micropores in shale gas reservoirs, i.e. viscous flow, slippage effect and Knudsen diffusion. This processing procedure has been widely adopted in the study of gas transporting mechanisms and numerical simulations in shale gas reservoirs since then. However, two problems need to be considered here before its further application. Firstly, Knudsen diffusion was employed to demonstrate the linear relationship between flow rate and pressure drop with constant diffusion coefficient and negligible viscous effects in nanopores. The reasonability of employing slippage viscous flow and Knudsen diffusion simultaneously in Eq. (55) is worth to be considered, whether it improves the simulation accuracy or complicates the problems. Besides, the physical basis of the procedure also needs to be verified. Secondly, Javadpour et al. [11] pointed out that the Knudsen diffusion supported the slip boundary at the nanopore surfaces, but the slippage phenomenon was again modified by correction factor F in Eqs. (55) and (56). Thus, we need to consider that whether the slippage phenomenon was repeatedly calculated by multiplying a correction factor F and adding Knudsen diffusion simultaneously.

(2) Weighted superposition [115]

Linear superposition of viscous/slip flow and Knudsen diffusion cannot cover the flow regime spectrum, especially for transition flow regimes [116]. Adzumi [112] introduced a weighting coefficient for free molecular flow, and pointed out that the coefficient increases smoothly from 0.7 to 1.0 as pressure decreases at low pressure conditions. Unfortunately, no formula or analytical modeling was proposed in his study. Based on the work of Adzumi [112], Liu et al. [117] defined a weighting coefficient for Knudsen diffusion according to the ratio of corresponding areas, which can be seen in Table 6. Rahmanian et al. [116] proposed a weighting factor expression as a function of Knudsen number (also in Table 6), where the model was applied to compare with DSMC data and produced good fitting results.

Taking the ratio of gas-gas molecules collision frequency to the total collision frequency as the weighting factor of slip flow (viscous flow and slippage effect) and taking the ratio of wall-gas molecules collision frequency to total collision frequency as weighting factor of Knudsen diffusion, Wu et al. [12] constructed a bulk phase gas transport model, with two weighting coefficients given as

$$\omega_v = \frac{1}{1 + Kn} \quad (57)$$

$$\omega_k = \frac{Kn}{1 + Kn} \quad (58)$$

In order to get the transport model in all flow regimes, Shi et al. [111] introduced a weight coefficient to accommodate the weighting between slip flow and Knudsen diffusion, which can be expressed as

$$\omega_v = \frac{1}{1 + \left(\frac{Kn}{Kn_{0.5}}\right)^{n_s-1}} \quad (59)$$

where $Kn_{0.5}$ and n_s are two constant, obtained by fitting with the model of Sakhae-Pour and Bryant [82] in transition flow regime. In their study, the two coefficients are obtained as $Kn_{0.5} = 4.5$ and $n_s = 5$.

The variance of weighting factor with Knudsen number was shown and compared in Fig. 21. With Knudsen number increasing from 0 to 0.5, the weighting factor of Knudsen diffusion increases and the weighting factor of viscous/slip flow decreases continuously. For Knudsen number higher than 0.5 in Liu et al. [117], weighting factor of slip/viscous flow increases and weighting factor of Knudsen diffusion decreases with Knudsen number increasing, which is not physically reasonable and becomes invalid. Similarly, the model of Rahmanian et al. [116] becomes invalid when Knudsen number is larger than 0.5, since the weighting coefficient for viscous flow become negative. The models of Wu et al. [12] and Shi et al. [111] are physically reasonable for all Knudsen numbers, where the weighting factor for slip flow decreases and weighting factor for Knudsen diffusion increases continuously with Knudsen number increasing, as we can see from Fig. 21(a) and Fig. 21(d) respectively. The difference of weighting factors from different models are very obvious, as we can see from Fig. 21(e) and Fig. 21(f). Therefore, the physical meaning and variance laws of weighting factors need further verification and investigation.

After obtaining the weighting factors of viscous/slip flow ω_v and of Knudsen diffusion ω_k , the apparent bulk phase gas transporting permeability can be expressed as

$$k_{\text{bulk}} = \omega_v Fk_{\text{ins}} + \omega_k k_{\text{Knudsen}} \quad (60)$$

9.1.2. Single slippage model

Single slippage model in this paper refers to those free gas transport models, where slippage effect or Knudsen diffusion play the same role for gas transporting in tight porous media. In other words, Knudsen diffusion is the reason for the occurring of slippage phenomenon and only one of them is considered in modeling. Yu et al. [2] mentioned that gas-wall collisions of Knudsen diffusion

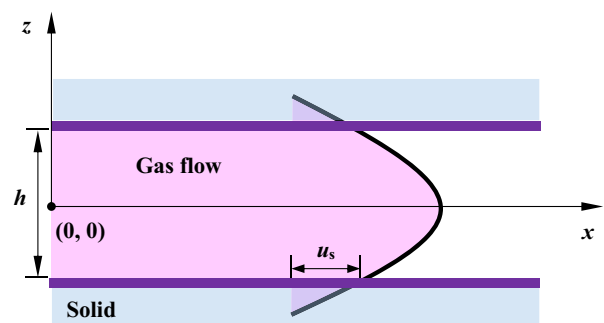


Fig. 22. Schematic of slip flow in two parallel plates.

induced the slip velocity at the pore surface, i.e. slippage effect and Knudsen diffusion are equivalent and describe the same mechanism, while gas–gas collisions are in accordance to viscous flow. Thus, the slippage effect should be considered by either of the following equations, where the slippage modification and Knudsen diffusion should not be repeatedly calculated.

$$k_{bulk} = Fk_{ins} \quad (61)$$

$$k_{bulk} = k_{ins} + k_{Knudsen} \quad (62)$$

The superposition of viscous flow and Knudsen diffusion in Eq. (62) is linear, with weighting factor for both mechanisms equaling to unity. From this standpoint, weight superposition method introduced above can also be applied in this section for the consideration of viscous flow and Knudsen diffusion simultaneously, i.e.

$$k_{bulk} = \omega_v k_{ins} + \omega_k k_{Knudsen} \quad (63)$$

Based on first order correlation of slip flow regime, Michel et al. [56] derived an apparent permeability as a superposition of viscous flow and Knudsen diffusion for ideal gas

$$k_{slip} = k_{ins} + \frac{3\pi}{16} \frac{\mu D_k}{p} \quad (64)$$

Comparing Eq. (64) to Eq. (48), we can see that slippage permeability is obtained by a superposition of viscous flow and Knudsen diffusion, with constant weighting factor 1 for viscous flow and $3\pi/16$ for Knudsen diffusion. Note that it was $3\pi/18$ in the reference [56], which was different from our derivation result in this review. The result can be seen as a Klinkenberg correction to permeability for gases flowing at relative low densities. Thus, Eq. (64) can also be transformed into a form of dynamic slippage factor. However, the first order approximations of the volumetric flux are not valid for the full range of slip flow regime [56].

Yu et al. [2] employed B-K correlation on intrinsic permeability to modify Knudsen diffusion of free gas, and the Knudsen diffusion part can be expressed as $(F-1) \cdot k_{ins}$, which is also the contribution of slippage effect according to the definition of single slippage model. A same procedure was also taken by Song et al. [22] Knudsen diffusion was first introduced into modeling gas flow in tight porous media by Ertekin et al. [87] when studying gas slippage effect. Knudsen diffusion was modelled by Fick's law under concentration field, and viscous flow was modelled by pressure field, assuming the two driving mechanisms exerted by pressure field and concentration field acted in parallel, i.e. linear superposition. The derived dynamic gas slippage factor considering dual driving fields was as below

$$b_k = \frac{p D_k C_g \mu}{k_{ins}} \quad (65)$$

Associating with Eq. (36) and comparing the dual-mechanism slippage factor b_k of Eq. (65) with equivalent permeability of Knudsen diffusion for real gases in Eq. (52), we can find that they are the same in essence.

9.2. Knudsen number based modification function

In this type of models, the apparent permeability of bulk phase gas transport in porous media is expressed as a function of Knudsen number

$$k_{bulk} = k_{ins} f(Kn) \quad (66)$$

where k_{bulk} is apparent permeability to describe bulk phase gas transport in tight porous media; k_{ins} is intrinsic permeability of tight porous media; $f(Kn)$ is modify function for rarefaction effects.

Beskok and Karniadakis [71] developed a unified model for ideal gas flow in micro-tubes, where this model was claimed to be valid for entire flow regimes

$$f(Kn) = (1 + \alpha Kn) \left(1 + \frac{4Kn}{1 - bKn} \right) \quad (67)$$

where α is dimensionless rarefaction coefficient; b is empirical slip coefficient, where $b = -1$ represents fully developed slip flow through tubes regardless of gas type.

Note that, the characteristic length was taken as the capillary radius when calculating Knudsen number Kn in Eq. (67), while the capillary diameter was adopted in our review. For unification, Eq. (67) was rewritten into the following form, where the characteristic length was given the value of pore diameter

$$f(Kn) = (1 + 2\alpha Kn) \left(1 + \frac{8Kn}{1 - 2bKn} \right) \quad (68)$$

The rarefaction coefficient α can be determined empirically by the following equation

$$\alpha = \frac{64}{3\pi(1 - 4/b)} \frac{2}{\pi} \arctan(4Kn^{0.4}) \quad (69)$$

Later, dimensionless rarefaction coefficient α is correlated empirically by Civan [118] as

$$\frac{\alpha_0}{\alpha} - 1 = \frac{A}{Kn^B}, A > 0, B > 0 \quad (70)$$

where α_0 is an asymptotic limit value; A and B are empirical constants. Civan [118] correlated a set of experimental data and obtained the value of $\alpha_0 = 1.358$, $A = 0.1780$ and $B = 0.4348$. Here, Eq. (70) is used as an alternative way of determining rarefaction coefficient α , which is mathematically simple and can provide satisfactory results. However, fitting procedures need to be conducted each time for specific experimental data.

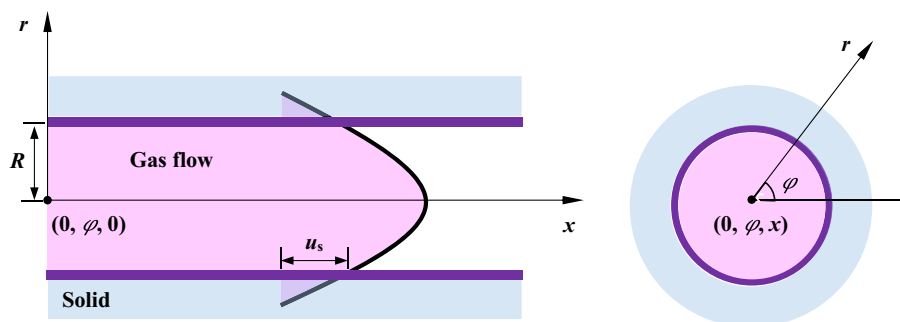


Fig. 23. Schematic of gas slippage flow in a micro-tube.

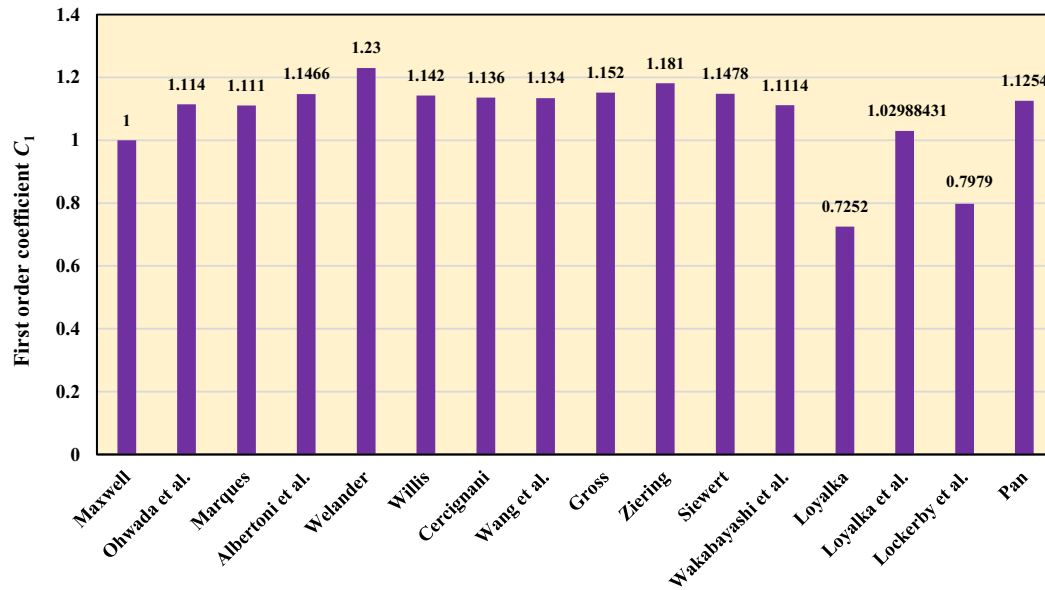


Fig. 24. Comparison of the first order slip coefficient C_1 from different studies.

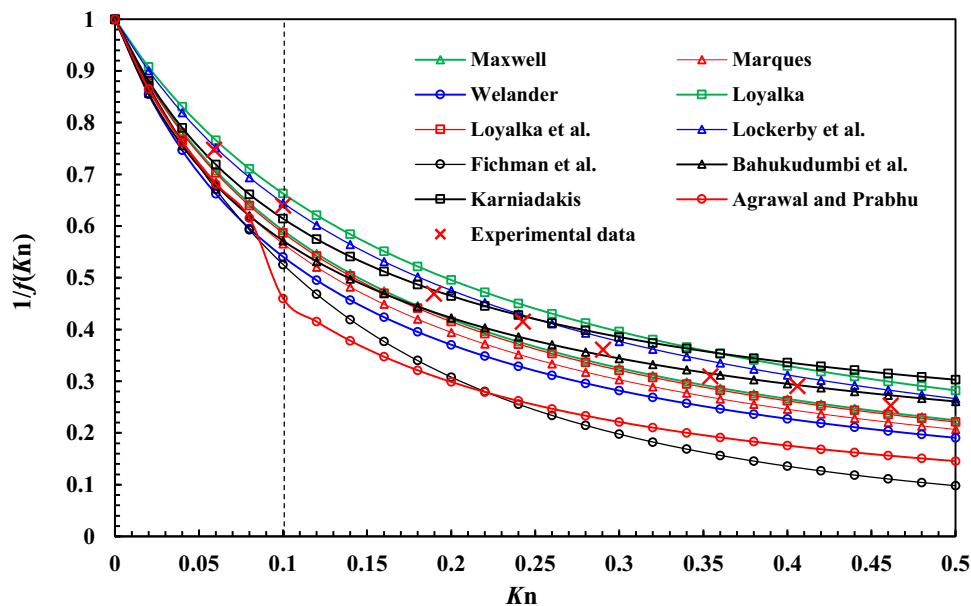


Fig. 25. The inverse of modification function with first order slip coefficients and their comparison with experimental data.

For gas flow in two parallel plates or a duct with rectangular cross section, the modification function can be expressed in the following form

$$f(Kn) = (1 + \alpha Kn) \left(1 + \frac{6Kn}{1 - bKn} \right) \quad (71)$$

where the characteristic length was taken height h between two parallel plates.

More generally, the modification function $f(Kn)$ can be derived from the slip boundary conditions introduced in 7.2 Slip boundary conditions. As we can see, the expression form of the modification function $f(Kn)$ for micro-tube with circular cross section and two parallel plates or a duct with rectangular cross section are different, while it is the same for two parallel plates and a duct with rectangular cross section. Thus, in the following section, we will

derive the modification function for two parallel plates and a micro-tube respectively.

9.2.1. Gas flow in microchannel

The modification function $f(Kn)$ for two parallel plates or a duct with rectangular cross section (Fig. 22) can be expressed as

$$f(Kn) = 1 + 6C_1Kn + 12C_2Kn^2 \quad (72)$$

where C_1 and C_2 are model parameters and differ in different model, which can be seen in 7.2 Slip boundary conditions. Detailed derivation of Eq. (72) can be seen in the Appendix A: derivation of $f(Kn)$ in paths with different shapes.

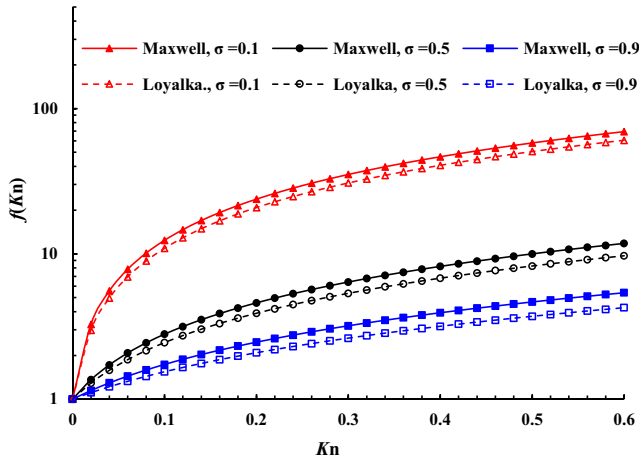


Fig. 26. The effects of TMAC on modification function $f(Kn)$.

9.2.2. Gas flow in micro-tube

The modification function for gas flow in a long micro-tube with circular cross section (Fig. 23) can be written as

$$f(Kn) = 1 + 8C_1Kn + 16C_2Kn^2 \quad (73)$$

where C_1 and C_2 are model parameters and differ in different model, which can be seen in 7.2 Slip boundary conditions and the detailed derivation of Eq. (73) can also be seen in the Appendix A: derivation of $f(Kn)$ in paths with different shapes.

9.2.3. Pseudo Knudsen number

In all above Knudsen number Kn based models, Knudsen number cannot be measured by direct laboratory measurement. In order to remedy this problem, Florence et al. [73] proposed a pseudo Knudsen number conception, which was correlated to measurable parameters of intrinsic permeability k_{ins} , absolute pressure p and porosity ϕ using non-linear regression, similar to the solving procedure of slippage factor b_k by Heid [88], Jones and Owens [37] or Sampath and Keighin [89]. The pseudo Knudsen

number is defined as Eq. (74), where a_2 , a_3 and a_4 are parameters obtained according to the regression results of studied reservoir dataset. Thus, similar notation need to be alerted that the dataset of k_{ins} , p and ϕ need to be sufficient and taken from the same depositional sequence. In another words, the coefficients of a_2 , a_3 and a_4 in Eq. (74) varies from one reservoir to another

$$Kn_p = a_2 \frac{1}{p} k_{ins}^{a_3} \phi^{a_4} \quad (74)$$

In Florence et al. [73] study, corresponding results were obtained as $a_2 = 2.62$, $a_3 = -0.553$ and $a_4 = 0.3897$ for Lower Cotton Valley data. After obtaining the relationship of Eq. (74), the pseudo Knudsen number Kn_p can be substituted into Knudsen number based models to calculate apparent permeability of bulk phase gas transport in tight porous media.

10. Model comparison

10.1. Knudsen number based modification function $f(Kn)$

10.1.1. First order slip model

Parts of the first order slip coefficients C_1 in Table 9 are drawn and compared in Fig. 24, whose values distributes between 0.7252 and 1.23. Especially, most values are in the range of 1.17 ± 0.06 , except those from Maxwell, Loyalka [119], Loyalka et al. [120], Lockerby et al. [95] and Pan [81], which include smaller value of the first order slip coefficient C_1 . Among these slip coefficients, the model of Welander [121] with slip coefficient $C_1 = 1.23$ produces the largest modification function $f(Kn)$ in Fig. 24, generating a largest slippage permeability for shale gas transporting system.

As we can see from Fig. 25, no model can match the experimental data well at both low and high Knudsen number conditions. The model of Lockerby et al. [95] match the experimental well in slip flow regime, while the model of Bahukudumbi et al. [122] fits the experimental data at a relatively high Knudsen number condition ($0.28 < Kn < 0.44$). Overall, the fitting performance of the first order slip models is poor since no model can match experimental data well at a large Knudsen number range. From this standpoint,

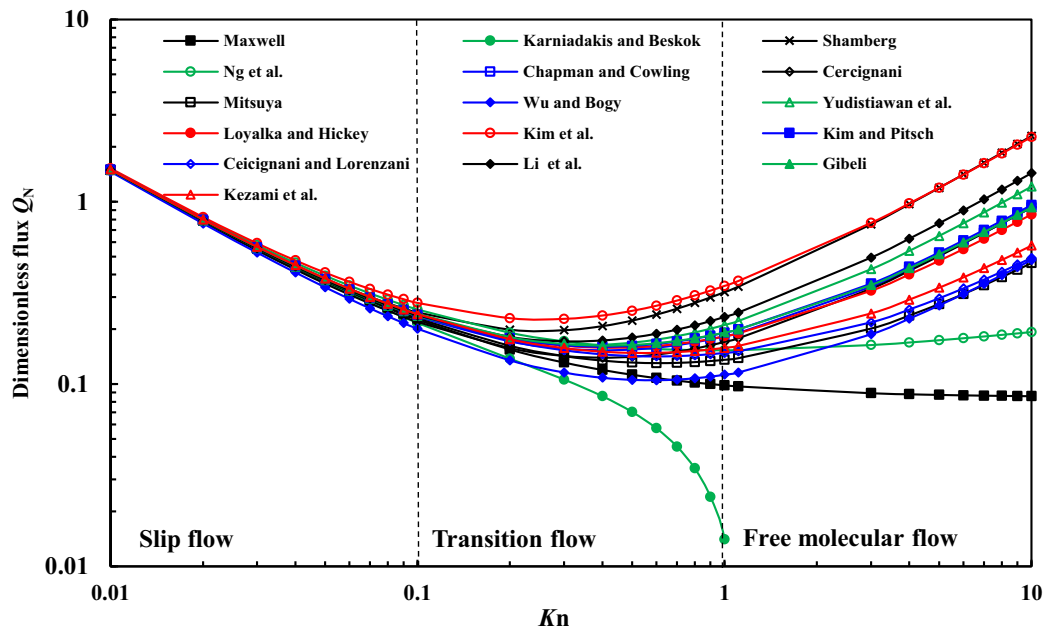


Fig. 27. The variance of dimensionless flow rate Q_N with Knudsen number Kn .

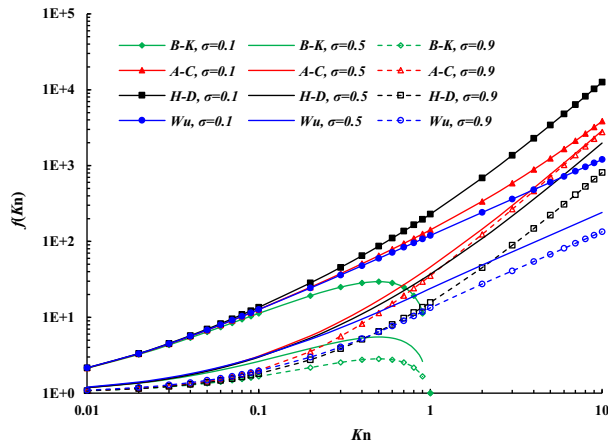


Fig. 28. The effects of TMAC on modification function of different models.

more accurate slip models are in urgent need to be developed to for gas transporting description with large Knudsen number.

TMAC affects the modification function severely, as we can see from Fig. 26. With TMAC increasing, the modification function $f(Kn)$ shows an inverse relation decline, implying a weak gas transporting capability. This is because TMAC represents the percentage of fully diffusive molecules to total molecule amount. Bigger TMAC value produces severer irregularity of molecules moving direction, which weakens the gas transporting capability along the moving direction. Though Loyalka model [119] produces the smallest slip coefficient C_1 in all first order slip models, the difference of modification function $f(Kn)$ between Maxwell model and Loyalka model is still small comparing to the variance of TMAC. Thus, it is an urgent and significant task to determine the TMAC efficiently and accurately either by experiments or by molecular dynamics. Though numerous TMAC values for simple gases and solid materials are available in the literature, the determination of TMAC for shale gas system is not straight and very challenging [16,109]. Studies have shown that TMAC is sensitive to gas–solid interface

conditions. Unfortunately, the effects of gas species, surface materials, surface temperature, porous media properties, and other surface conditions on TMAC are still not well understood up to date [84]. Empirically, TMAC is usually taken as unity or slightly smaller than unity for most of the practical engineering conditions, which was also indicated and verified by experimental measurement results for most of the TMAC values ranging from 0.74 to 0.94 [84].

10.1.2. Second order slip model

The second order slip models in Table 10 for fully diffusive reflection are evaluated in the form of dimensionless flow rate versus Knudsen number Kn , as is shown in Fig. 27. The dimensionless flow rate is defined as [123]

$$Q_N = \frac{\sqrt{\pi}(1 + 6C_1Kn + 12C_2Kn^2)}{126Kn} \quad (75)$$

As we can see and predict, all models produce nearly the same dimensionless flow rate when Knudsen number is smaller than 0.01. With Knudsen number increasing, small discrepancies occur at late slip flow regime among different models. The discrepancies magnify and polarize in transition flow regime, where the models of Lockerby and Reese [124] and Karniadakis and Beskok [125] with negative second order slip coefficients produce a obvious downward trend, while other second order slip models show an upward tendency. During free molecular flow regime, the dimensionless flow rate of most second order slip models keep the upward trend with similar speed, except the model of Ng et al. [126], which mainly keeps the same during this flow regime. Among the second order slip models in Fig. 27, the models of Kim et al. [127] with $C_1 = (6/\pi)^{0.5}$ and $C_2 = 4/\pi$ and Shamberg [128] with $C_1 = 1$ and $C_2 = 5\pi/12$ generate similar and two largest prediction results in free molecular flow regime. The model of Wu and Bogoy [129] with $C_1 = 2/3$ and $C_2 = 1/4$ predicts the smallest dimensionless flow rate for Knudsen number $1 < Kn < 2$ and Ng et al. [126] predicted the smallest for $Kn > 2$. The difference among these models may be caused by different assumptions during theoretical derivation or different manipulating procedures during

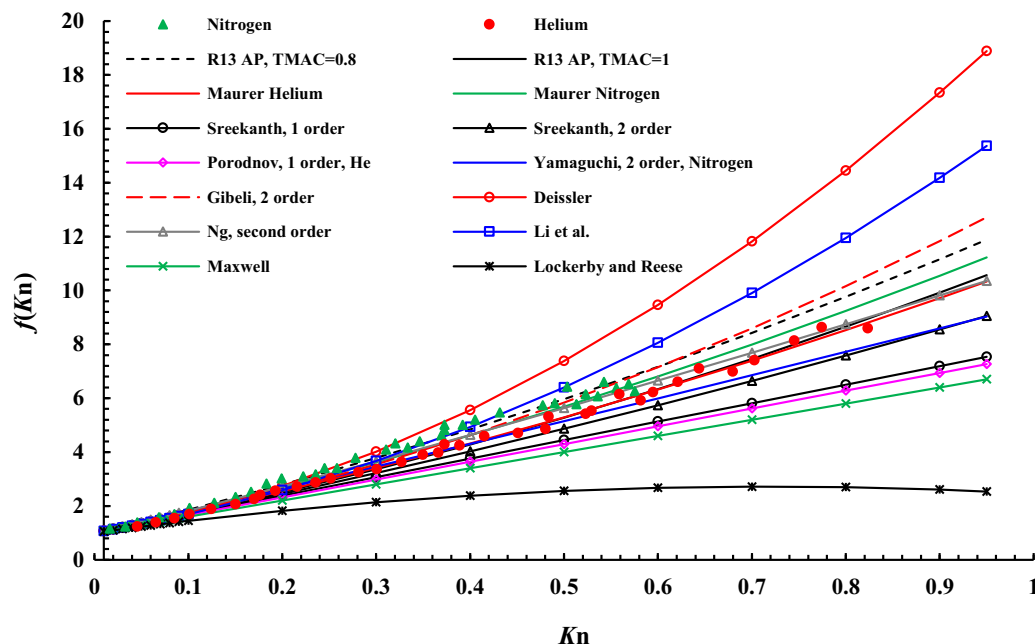


Fig. 29. Comparisons of different slip models and experimental data in different Knudsen number conditions.

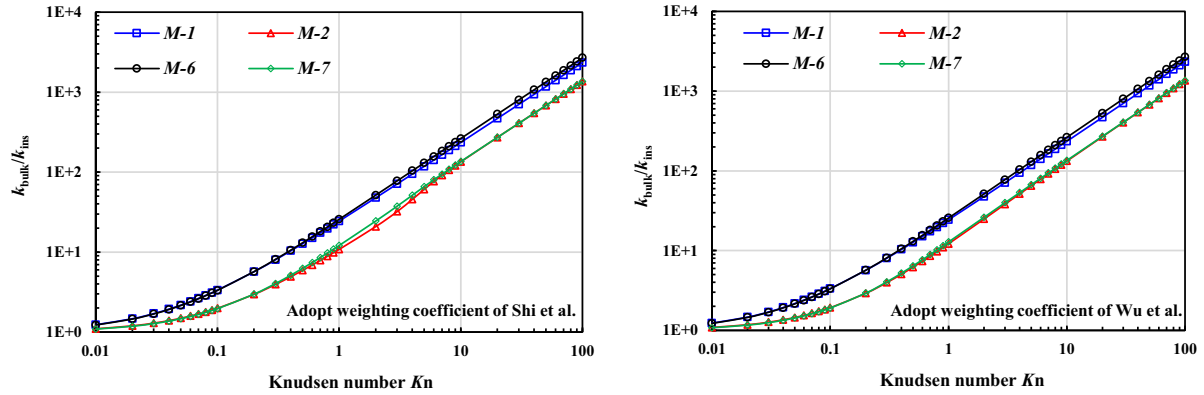


Fig. 30. The comparison of double slippage models as well as with different weighting coefficient.

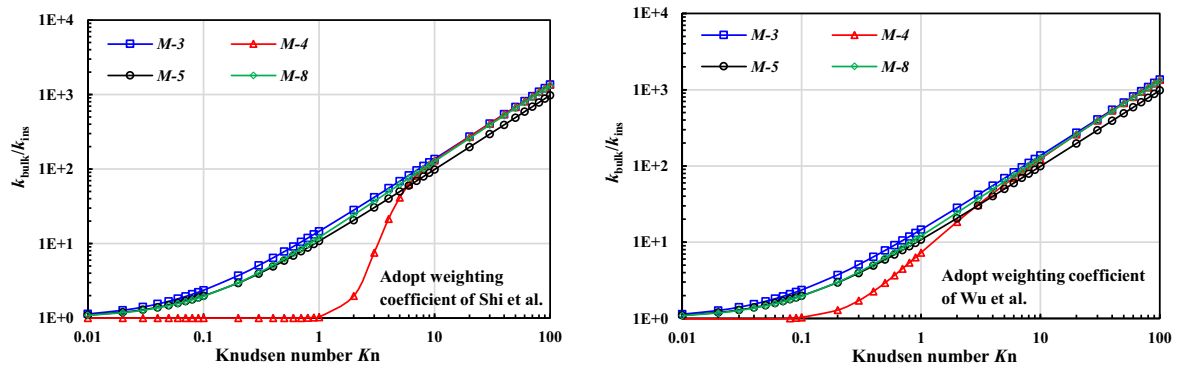


Fig. 31. The comparison of single slippage models as well as with different weighting coefficient.

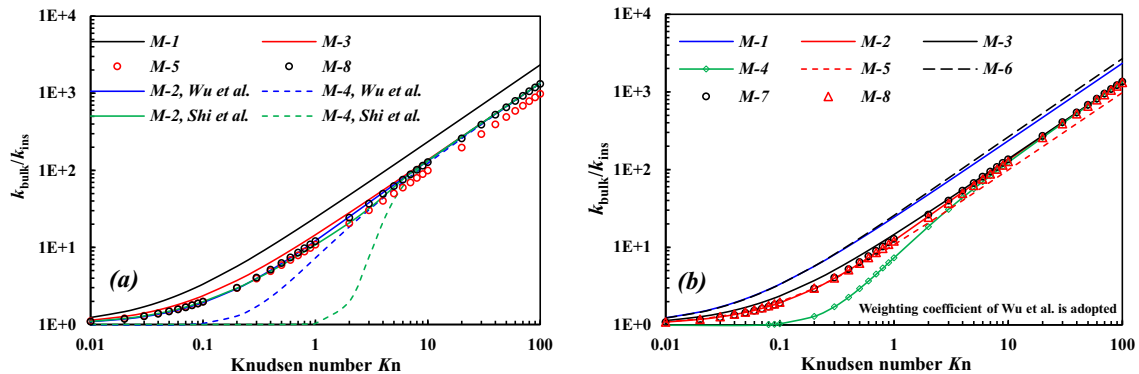


Fig. 32. The variance of $k_{\text{bulk}}/k_{\text{ins}}$ versus Knudsen number Kn for different models.

experiments. Overall, it is not appropriate to take a negative value for second order slip coefficient, since the predicted dimensionless flow rates are below zero in high Knudsen number conditions.

As we mentioned, the first and second order slip coefficients may relate to TMAC, if the reflection are not fully diffusive. Here, four models of Beskok and Karniadakis [71] (B-K for short), Aubert and Colin [130] (A-C for short), Hsia and Domoto [131] (H-D for short) and Wu [129] with TMAC equaling to 0.1, 0.5, and 0.9 were studied and compared, as shown in Fig. 28. For TMAC = 0.1, the prediction from Beskok and Karniadakis [71] generates the largest results, while the model of Aubert and Colin [130] generates the largest for TMAC = 0.5 and 0.9. Thus, the second order slip coefficients with TMAC in it implies a larger gas

transporting capacity when the TMAC value is small, while the model without TMAC in second order slip coefficients dominate the flow when TMAC is a large value. The prediction results of A-C [130] and Wu [69] for TMAC = 0.1, H-D [131] and Wu for TMAC = 0.9 are comparable respectively for Knudsen number ranging from 0.01 to 1. The model of Beskok and Karniadakis [71] produces a decrease at the Knudsen number near 0.6 and becomes negative values in free molecular flow regime, since the second order slip coefficient is negative and second order slip coefficient dominate the flow in high Knudsen number conditions. From this standpoint, we can deduce that the negative second order slip coefficient is not appropriate for rarefied gas effect description in nanopores.

In Fig. 29, the first and second order slip coefficients C_1 and C_2 obtained from experimental measurement in Table 12 were taken into the modification function of Eq. (72) to compare with each other as well as the experimental data. Some theoretical derived slip models were also included in Fig. 29. As we can see, the discrepancies between different models as well as experimental data at small Knudsen number range ($Kn < 0.1$). With Knudsen number increasing, the differences between different models become more and more obvious. All second order models produce larger results than those from Maxwell model [94] except Lockerby and Reese [124]. This is because the second order slip coefficient C_2 in Lockerby and Reese's study [124] is a negative value, while all others in Fig. 29 are positive. The slip models from the studies of Maxwell [94], Porodnov [132], Sreekanth [97] underestimates the modification function for Knudsen number larger than 0.2, while the second order slip model from Yamaguchi [101] fits the data well until Knudsen number reaches 0.52. Overall, the models of Maurer et al. [100], Kazemi et al. [7], Ng et al. [126], Gibeli model [133] can fit the experimental data on a relatively good performance, while Li et al. model [134] and Deissler model [96] severely overestimates the results at large Knudsen numbers.

10.2. Bulk phase gas transporting models

10.2.1. Model classification

As we mentioned, superposition models of viscous/slip flow and Knudsen diffusion can also be deemed as the first order approximation. Though a large number of bulk phase gas transporting models were established, they can be clarified into the following eight types after derivation and comparison, mathematically expressed as ① $F \cdot k_{ins} + k_{knudsen}$; ② $\omega_v \cdot F \cdot k_{ins} + \omega_k \cdot k_{knudsen}$; ③ $k_{ins} + k_{knudsen}$; ④ $\omega_v \cdot k_{ins} + \omega_k \cdot k_{knudsen}$; ⑤ $F \cdot k_{ins}$; ⑥ $f(Kn) \cdot k_{ins} + k_{knudsen}$; ⑦ $\omega_v \cdot f(Kn) \cdot k_{ins} + \omega_k \cdot k_{knudsen}$; ⑧ $f(Kn) \cdot k_{ins}$. First, we will derive the former six models into the form of Knudsen number for convenience of comparison. For simplicity, the equivalent permeability of Knudsen diffusion adopts the ideal gas expression in Eq. (48), and the intrinsic permeability is obtained by Hagen-Poiseuille model for gas flow in a micro-tube. Then, the representative eight types of models are compared quantitatively.

10.2.2. Induction of various bulk phase gas transporting models

For double slippage model of linear superstition, the bulk phase gas transport permeability can be expressed as

$$k_{bulk} = Fk_{ins} + k_{knudsen} = k_{ins}(F + \frac{k_{knudsen}}{k_{ins}}) \quad (76)$$

Substituting the expression of Eqs. (40) and (48) and Hagen-Poiseuille permeability $k_{ins} = r^2/8$ into Eq. (90), Eq. (90) transforms into

$$k_{bulk} = k_{ins} \left[1 + 8 \left(\frac{2}{\sigma} - 1 \right) \left(\frac{\mu}{p \cdot 2r} \sqrt{\frac{\pi RT}{2M}} \right) + \frac{128}{3\pi} \left(\frac{\mu}{p \cdot 2r} \sqrt{\frac{\pi RT}{2M}} \right) \right] \quad (77)$$

Comparing Eq. (77) to the expression of Knudsen number in Eq. (32), we obtain the bulk phase gas transport permeability as

$$k_{bulk} = k_{ins} \left\{ 1 + \left[8 \left(\frac{2}{\sigma} - 1 \right) + \frac{128}{3\pi} \right] Kn \right\} \quad (78)$$

Taking the similar procedure as above for double slippage model of linear superposition, other models (② – ⑥) are also expressed in Knudsen number, as is shown in Table 7. The Knudsen number based modification function $f(Kn)$ was chosen as Beskok and Karniadakis model [71] in Eq. (68)

10.2.3. Model comparison

The eight type of models concluded in Table 7 almost represent all type of bulk phase gas transporting models in shale gas permeability study. In this section, different models will be compared to find out their differences and similarities, which is conducive to choose the proper model during in practical application.

In Fig. 30, the double slippage models of linear and weighted superposition are compared. Since the weighting coefficient for single slippage model in Table 6 become invalid in high Knudsen number conditions, the weighing coefficients of double slippage model, which is suitable to all flow regimes, are employed here for model comparison. The value of k_{bulk}/k_{ins} from linear superstition models is much higher than that from weighted superstition models. In linear superstition models, the Klinkenberg correction factor F produce almost the same results as that of Beskok and Karniadakis model [71] when Knudsen number is smaller than unity. The difference between the two models is also not significant in high Knudsen numbers. For weighted superposition models, the difference mainly locates in transition flow regime, which is more obvious if the weighting coefficients of Shi et al. [111] are adopted.

For single slippage models as in Fig. 31, the models of M-3, M-5 and M-8 generates similar results, while the M-4 model results show an obvious deviation with the others. This is because in single slippage models, Knudsen diffusion is deemed as the reason of gas slippage on pore surface in Knudsen number range of 0.001–0.1, while the weighting coefficients of Wu et al. and Shi et al. are for double slippage models, where Knudsen diffusion is employed to describe free molecular flow in Knudsen number range of $Kn > 10$. Thus, the type of M-4 model is not appropriate for gas transport description in shale gas reservoir system. M-3 model adopts Klinkenberg factor F and M-8 model adopts modification function $f(Kn)$ to account for the slippage effect or slip boundary condition. M-5 model considers the linear superposition of viscous flow and Knudsen diffusion, where Knudsen diffusion can be seen as a slip boundary in essence. Therefore, the results from models of M-3, M-5 and M-8 are close, the physical meaning of which are also the same. Comparing to M-8, M-3 overestimates the results when $Kn < 10$, while M-5 underestimates the results when $Kn > 1$.

Though weighted superposition of viscous flow and Knudsen diffusion (M-4) is not appropriate to describe gas transport in shale nanopores, the changing tendency is also included in Fig. 32 to examine the rationality of weighting factor from Shi et al. [111] and Wu et al. [12]. If the slippage effect is ignored and Knudsen diffusion is employed to describe gas migration in free molecular regime, the single slippage model based on weighted superposition of viscous flow and Knudsen diffusion should satisfy that viscous flow occurs when $Kn < 0.1$ (continuum flow) and it is purely Knudsen diffusion when $Kn > 10$ (free molecular flow). There should be a smooth transition from continuum flow (slippage effect is ignored) to free molecular flow. From this standpoint, we can see that the weighting coefficients of Wu et al. [12] produce more reasonable results. The double slippage model based on weighted superposition of slippage flow and Knudsen diffusion (M-2 model) yields similar results with single slippage model of M-5 and M-8. Specially, the results of weighted superposition (M-2 or M-7) of slippage flow and Knudsen diffusion using weighting coefficients of Wu et al. [12] completely coincide with that from M-8, which is the viscous flow corrected by modification function $f(Kn)$. The single slippage model based on linear superposition of viscous flow and Knudsen diffusion (M-3) also produces similar results, while the results of double slippage model based on linear superposition of slip flow and Knudsen diffusion (M-1 or M-6) are much higher. According to the comparison of eight type of models in Fig. 32 (b), we can find that the double slippage models based on weighted superposition of slip flow and Knudsen diffusion (M-2 and M-7)

match well with single slippage model corrected by modification function $f(Kn)$ (M-8) in all Knudsen number range. Therefore, these three methods are recommended to calculate bulk phase gas transporting capability in shale nanopores.

From the above analysis, conclusions could be drawn that: (i) the physical meaning of linear superposition of viscous flow and Knudsen diffusion (M-3) is not considering viscous flow and Knudsen diffusion simultaneously, but a kind of Klinkenberg correction for slippage effect; (ii) the results of double slippage model based on weighted superposition of slippage flow and Knudsen diffusion (M-2 and M-7) yields similar results with that of single slippage model (M-3 and M-8); (iii) the single slippage model based on weighted superposition of viscous flow and Knudsen diffusion (M-4) is not appropriate to describe gas migration in shale nanopores, the results of which is obviously low. Note that, this conclusion is based on the weighting coefficients of double slippage model in Table 6. If the weighting coefficients of single slippage model in Table 6 are used, M-4 can also describe gas transport in shale nanopores with Knudsen number smaller than 0.5; (iv) double slippage model based on linear superposition of slip flow and Knudsen diffusion (M-1 and M-6) severely overestimates the results, where the effects of gas slippage and Knudsen diffusion are re-considered; (v) the single slippage model of viscous flow corrected by Klinkenberg factor F (M-5) underestimates the results in transition and free molecular flow regime; (vi) the double slippage models based on weighted superposition of slip flow and Knudsen diffusion (M-2 and M-7) and the single slippage model corrected by modification function $f(Kn)$ (M-8) yield similar result in all Knudsen number range, with only slightly difference in transition flow regime, which are recommended to be used in bulk phase gas transporting capability calculation in further investigation; (vii) Weighted superposition of slip flow and Knudsen diffusion (M-2 and M-7) can be used to differentiate different flow mechanisms in total flux, while single slippage model corrected by modification function $f(Kn)$ (M-8) is used to calculate total bulk phase gas transporting capacity in nanopores.

11. Surface diffusion

One of the main differences between flow behaviors in inorganic matrix and organic nanopores is that surface diffusion may occur in adsorbed gas layer during production, which is a key issue in predicting gas production in shale gas reservoirs [9]. Surface diffusion is driven by concentration gradient of adsorbed gas, and different models are proposed to describe this type of gas transport in organic nanopores.

11.1. Surface diffusion modeling

There are mainly two methods to establish surface diffusion model. One is to couple Fick's law with Langmuir adsorption theory, and the other is through Maxwell-Stefan method with chemical potential.

11.1.1. Fick's law with Langmuir adsorption

Fick's law is usually employed to describe surface diffusion, since it is driven by concentration gradient. The mass flux J_{surface} of surface diffusion can be written in the following form as [14,17,50,135,136]

$$J_{\text{surface}} = \rho_a u_a = -D_s \nabla C_s \quad (79)$$

where J_{surface} is surface diffusion flux; ρ_a is adsorbed gas density; u_a is adsorbed gas velocity; D_s is surface diffusion coefficient; C_s is adsorbed gas concentration, $\text{kg}(\text{gas})/\text{m}^3(\text{solid})$.

Adsorbed gas density can be calculated by the methods introduced in Part 2.4.2. In the study of Yu et al. [9], $(2.16 \cdot \rho_{\text{free}})$ is chosen as adsorbed gas density ρ_a , i.e. $\rho_a = 2.16 \times \rho_{\text{free}}$.

According to the study in Part 2, adsorbed gas concentration can be expressed as

$$C_s = \rho_b G_a \quad (80)$$

Adopting Langmuir adsorption isotherm in Eq. (80) and substituting it into Eq. (79), the mass flux of surface diffusion transforms into

$$J_{\text{surface}} = -D_s \rho_b G_m \frac{bp/Z}{(1 + bp/Z)^2} c_g \nabla p \quad (81)$$

Comparing Eq. (81) to Darcy's law of Eq. (35), the equivalent permeability of surface diffusion k_{surface} can be written as

$$k_{\text{surface}} = \mu_{\text{free}} D_s \frac{G_m \rho_b}{\rho_g} \frac{bp/Z}{(1 + bp/Z)^2} c_g \quad (82)$$

here, in order to be unified with Darcy flux equation, underground gas density ρ_g , rather than adsorbed gas density ρ_a is adopted, which need to be differentiated with the expression form in Eq. (79).

11.1.2. Maxwell-Stefan method with chemical potential

In this method, the driving force of surface diffusion is supposed to be a chemical potential gradient [12,45,49,137], and the surface diffusion flux can be expressed as

$$J_{\text{surface}} = -L_m C_s \nabla \psi \quad (83)$$

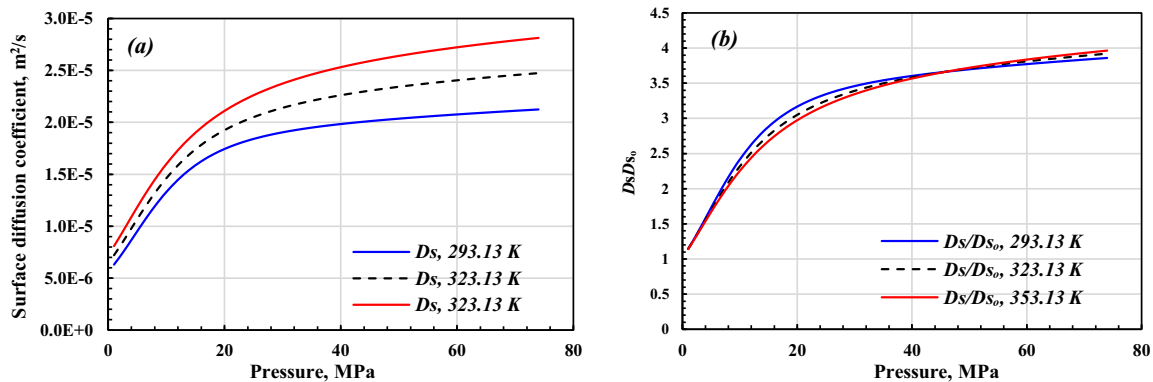


Fig. 33. Surface diffusion coefficient D_s (a) and the ratio of D_s to D_{s0} (b) at different pressure and temperature conditions.

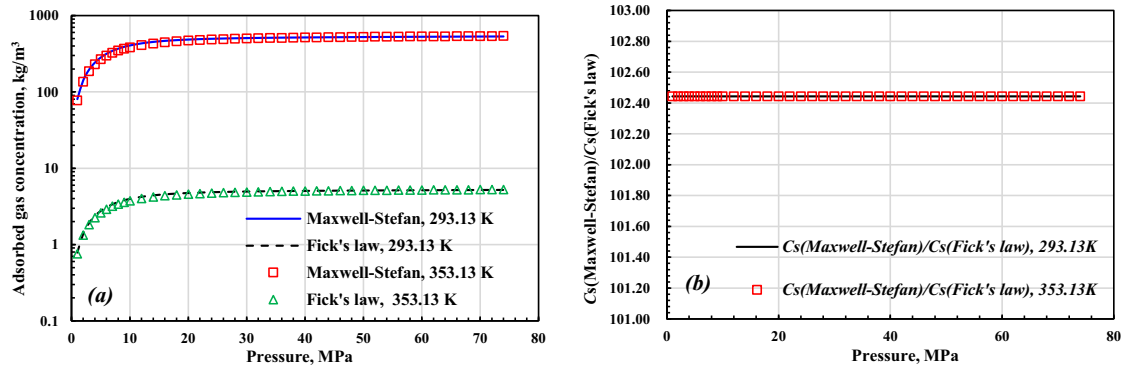


Fig. 34. Adsorbed gas concentration (a) and their ratios (b) at different pressure and temperature conditions.

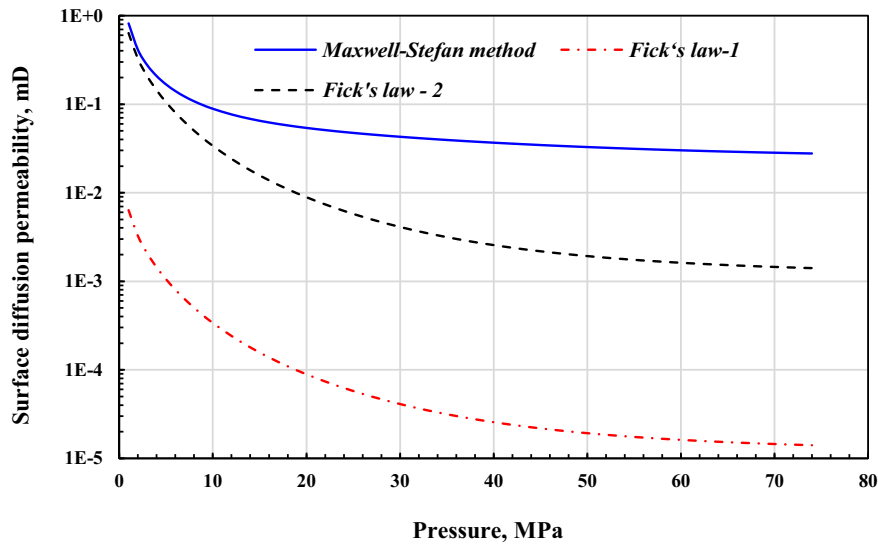


Fig. 35. Surface diffusion permeability from Maxwell-Stefan model Eq. (87) and Fick's law model Eq. (82).

where L_m is gas mobility, mol·s/kg; ψ is chemical potential, J/mol; L_m and ψ can be calculated by

$$L_m = \frac{D_s}{RT} \quad (84)$$

$$\psi = \psi_0 + RT \ln p \quad (85)$$

$$C_s = \frac{4M\theta}{\pi d_m^3 N_A} \quad (86)$$

Substituting Eqs. (84) and (85) into Eq. (83), we can obtain the equivalent permeability of surface diffusion after some mathematical manipulation

$$k_{\text{surface}} = \frac{\mu_{\text{free}} D_s}{p} \frac{1}{\rho_g} \frac{4M}{\pi d_m^3 N_A} \left(\frac{bp/Z}{1 + bp/Z} \right) \quad (87)$$

Note that, the physical meaning of adsorbed gas concentration C_s in Eqs. (79) and (83) is different. It refers to gas mass in unit solid volume in Eq. (79), while it refers to gas mass in unit adsorbed gas volume in Eq. (83). This difference leads to great difference between surface diffusion permeability between the two models, which will be discussed later.

11.1.3. Considering porous media properties

Considering surface diffusion in organic porous media, the permeability enhancement due to surface diffusion was calculated by Naraghi et al. [138] in an equivalent form as the following equation

$$k_{\text{surface}} = \mu_{\text{free}} D_s \frac{G_m \rho_b}{\rho_g} \frac{(1-\phi)}{\text{TOC}} \frac{\phi}{\tau} \frac{1}{b(p + \frac{1}{b})^2} \quad (88)$$

where TOC is volumetric total organic carbon content; τ is tortuosity; ϕ/τ is a parameter to account for porous media properties, which can be removed for a single capillary.

11.1.4. Surface diffusion coefficient calculation

Surface diffusion coefficient in Eqs. (82), (87) and (88) can be calculated by [9,22]

$$D_s = D_{s0} \frac{(1-\theta) + \frac{\kappa}{2}\theta(2-\theta) + [\psi(1-\kappa)](1-\kappa)\frac{\kappa}{2}\theta^2}{(1-\theta + \frac{\kappa}{2}\theta)^2} \quad (89)$$

where $\psi(1-\kappa)$ is Heaviside step function; κ is the ratio the rate constant for blockage to the rate constant for forward migration, dimensionless and given the value of 0.5 in Wu et al. [12]; D_{s0} is surface diffusion coefficient at the surface coverage equaling to zero, which can be calculated as [4,22]

$$D_{s0} = 8.29 \times 10^{-7} T^{0.5} \exp\left(-\frac{\Delta H^{0.8}}{RT}\right) \quad (90)$$

where ΔH is the isosteric adsorption heat at the surface coverage θ equaling to zero and it was given the value of 16,000 J/mol in the study of Wu et al. [12].

Heaviside function $\psi(1 - \kappa)$ has the following property

$$\psi(1 - \kappa) = \begin{cases} 0, & \kappa \geq 1 \\ 1, & 0 \leq \kappa < 1 \end{cases} \quad (91)$$

where κ is the ratio of the blocking velocity coefficient κ_b to the forward velocity coefficient κ_m .

In addition, surface diffusion coefficient can also be measured by experiment, and some measured values are shown in Table 8.

11.2. Comparison of surface diffusion models

11.2.1. Parameters calculation

Surface diffusion coefficients at different pressure and temperature conditions are calculated, as shown in Fig. 33(a). If isosteric adsorption heat ΔH is chosen 16000 J/mol [12], surface diffusion coefficient is in the range of 6×10^{-6} m²/s to 2.8×10^{-5} m²/s. Surface diffusion coefficient is bigger at higher temperature conditions, while it increases fast with temperature increasing when pressure is smaller than 20 MPa. With temperature further increasing, surface diffusion increases with relatively slow speed, comparing to low pressure conditions. Meanwhile, the effects of gas coverage on surface diffusion coefficient values are evaluated in Fig. 33(b). With pressure increasing, gas coverage increases nonlinearly, and the ratio of surface diffusion coefficient considering gas coverage D_s to surface diffusion coefficient at gas coverage equaling to zero D_{s0} also increases in a Langmuir type. D_s can be more than 3.5 times of D_{s0} when pressure is over 40 MPa, implying gas coverage has big effects on surface diffusion coefficients and cannot be ignored. Comparing to pressure, temperature has slight effect on the ratio of D_s/D_{s0} , meaning the effects of gas coverage on surface diffusion coefficients at different temperature conditions are similar.

As mentioned, the adsorbed gas concentration in two models has different meanings. In Eq. (79) it refers to adsorbed gas mass in unit volume of solid, while in Eq. (83) it refers to adsorbed gas mass in unit volume of adsorbed gas, equivalent to adsorbed gas density. In both models, gas concentration increases fast with pressure increasing when pressure is smaller than 10 MPa, and nearly keeps constant when pressure is larger than 20 MPa, as shown in Fig. 34(a). This is because in both models, Langmuir adsorption is assumed, where gas adsorption increases fast at low pressure and nearly keeps constant at high pressure conditions, with pres-

sure increasing. The influence of temperature on adsorbed gas concentration can be ignored in both models, as shown in Fig. 34(a). In Fig. 34(b), adsorbed gas concentration calculated by Eqs. (80) and (86) are compared by taking their ratios. The ratios keep constant at different pressure and temperature conditions, with the value equaling to 102.44. From this point, we can say that the difference of adsorbed gas concentration from Eqs. (80) and (86) is due to the difference of reference volume. This difference directly leads to significant discrepancy of surface diffusion permeability between two models, as shown in Fig. 35. The adsorbed gas concentration calculated from Eq. (83) ranging from 77.41 kg/m³ to 538.58 kg/m³, while it ranges from 0.76 kg/m³ to 5.26 kg/m³ from Eq. (79). In future study, the expression needs to be unified for calculation and comparison. Note that, the upper limit of adsorbed gas concentration from Eq. (83) is much bigger than usual adsorbed gas density, which need to be verified for its accuracy.

11.2.2. Model comparison

In this part, the two surface diffusion models introduced above are compared by taking the same value of mutual parameter, such as gas deviation factor Z , pressure p , temperature T , gas coverage θ , surface diffusion coefficient D_s . Other parameters such as adsorbed gas concentration C_s and gas compressibility c_g are calculated by respective methods. For Fick's law-1 model in Fig. 35, maximum adsorbed capacity G_m is chosen the value of 0.0024 kg(gas)/kg(-solid), while in Fick's law-2 model, $G_m = 0.2459$ kg(gas)/kg(solid) is adopted, where a coefficient of 102.44 was used to unify two different expressions of adsorbed gas concentration, as shown in Fig. 34(b). As we can see from Fig. 35, without the unification of adsorbed gas concentration expressions, calculation results of surface diffusion permeability from Fick's law-1 model is two to three orders lower than that from Maxwell-Stefan model. Obviously, the difference is too significant to be tolerate. After the unification of adsorbed gas concentration, namely multiplying the adsorbed gas concentration from Eq. (80) by 102.44, the surface diffusion permeability from Fick's law-2 model becomes similar to that of Maxwell-Stefan model when pressure is smaller than 3 MPa. However, Fick's law-2 model decreases faster than Maxwell-Stefan model with pressure increasing. Therefore, this is because an extra term of $(1 + bp/Z)$ is in Fick's law model of Eq. (82) comparing to Maxwell-Stefan model of Eq. (87). In practical, it is in urgent need to find out which model is more appropriate to describe surface diffusion, considering the large discrepancies between two different models.

11.2.3. Surface diffusion role analysis

Surface diffusion is a unique gas transporting mechanism in organic shale nanopores, which has drawn great attention in recent

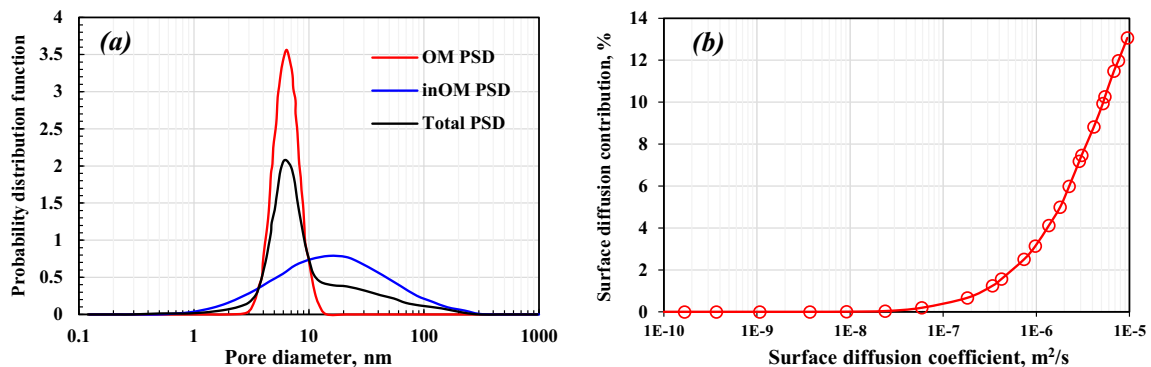


Fig. 36. Pore size distribution of shale samples under study (a) and surface diffusion contribution to total gas flow with different surface diffusion coefficient (b).

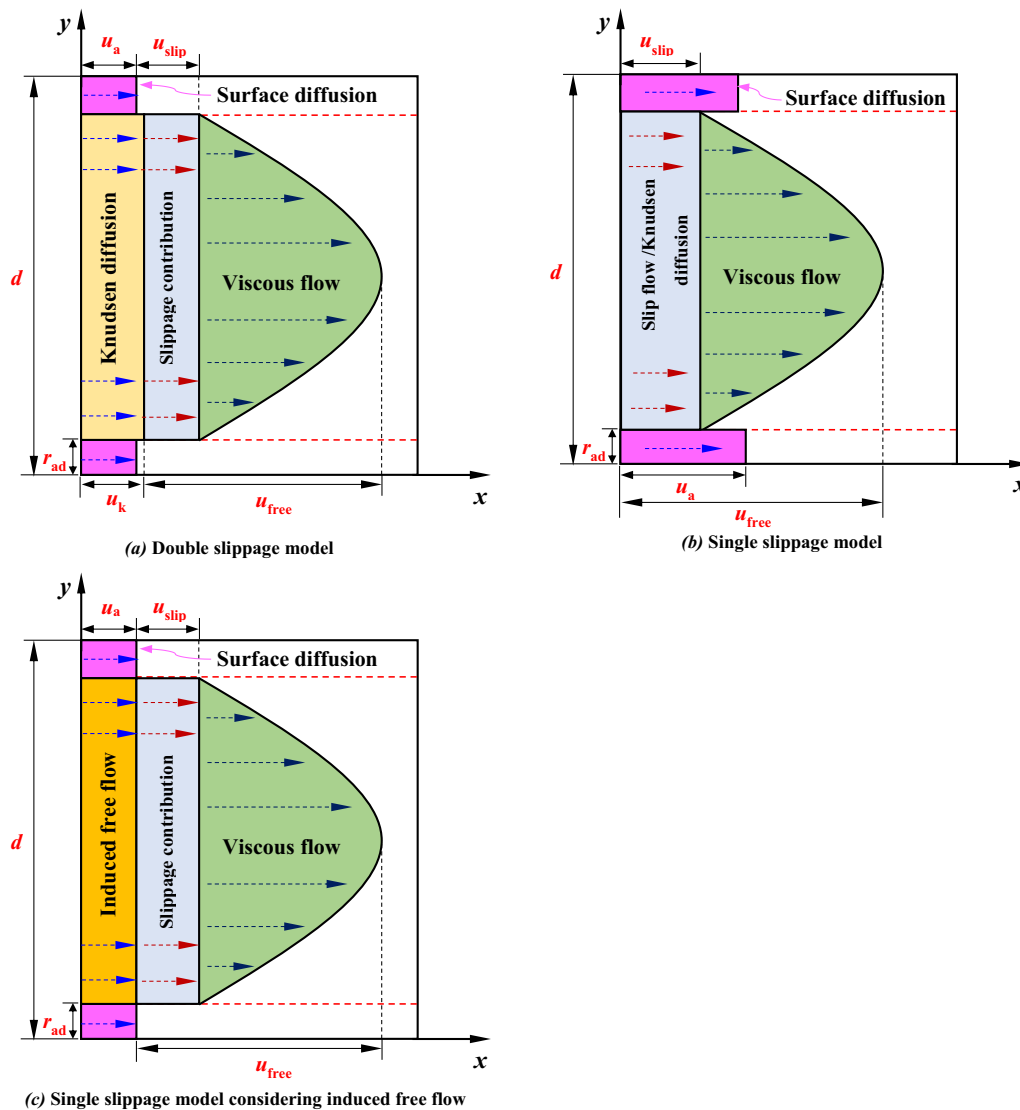


Fig. 37. Schematic of different transport mechanisms of shale gas in organic nanopores (revised from Yin et al. [9]).

years. However, the role of surface diffusion in total shale gas transporting process is still not clear, with two completely opposite kind of opinions obtained. Part studies pointed out surface diffusion is a significant gas transporting mechanism in shale gas reservoirs, since nanopores are developed and adsorbed gas is abundant, while others thought surface diffusion plays a limited role in gas transporting process and can be ignored.

Surface diffusion is considered to be an important gas transporting mechanism in shale gas reservoirs, considering the large contribution of surface diffusion to total gas transporting flux in small pores and nanopores are very developed in shale formations [12]. They pointed out that gas transfer is always dominated by surface diffusion over all the ranges of pressure in pores with diameter smaller than 4 nm, and it cannot be ignored under the condition of pore diameters ranging from 4 nm to 50 nm and pressure < 5 MPa [139]. Zhang et al. [17] proposed an apparent permeability model to account for slip flow, Knudsen diffusion and surface diffusion in shale gas reservoirs, and found that surface diffusion affected total flux seriously by accounting for 25% at low pressure, where surface diffusion cannot be neglected in gas transport study. Yin et al. [9] thought surface diffusion is supposed to enhance the free gas flow and generate the induced free gas flow

in bulk phase, while it is found that adsorbed surface diffusion may offer either the enhancement or impediment effects to bulk gas flow which is mainly determined by the magnitude of surface diffusion coefficient [136].

The simulation results of Wang et al. [50] indicated that, under most real shale gas reservoir conditions, the corrected permeability due to gas desorption and the non-Darcy flow permeability are dominant in the apparent permeability, while the surface diffusion permeability caused by adsorbed gas can be largely overlooked. Though artificially increasing the value of diffusion coefficient from $10^{-7} \text{ m}^2/\text{s}$ to $10^{-5} \text{ m}^2/\text{s}$ can lead to a significant contribution of surface diffusion to apparent permeability, the commonly accepted value of surface diffusion coefficient is in the order of $10^{-7} \text{ m}^2/\text{s}$ or smaller, as shown in Table 8. In the study of Naraghi et al. [138], the effect of surface diffusion on shale permeability was studied for diffusion coefficient ranging from $1 \times 10^{-20} \text{ m}^2/\text{s}$ to $1 \times 10^{-5} \text{ m}^2/\text{s}$, where the pore size distribution of the studied samples was shown in Fig. 36(a). Results showed that the contribution of surface diffusion is relatively negligible for surface diffusion coefficient values up to $1 \times 10^{-6} \text{ m}^2/\text{s}$, as can be seen from Fig. 36(b). Therefore, surface diffusion may not be an important transport mechanism in shale systems, since the typical values of

surface diffusion coefficient for shale samples are in the order of 10^{-20} m²/s as reported by Young et al. [55,138].

However, the values of surface diffusion coefficient differ significantly from different studies. Chen et al. [140] pointed out that Knudsen diffusion and surface diffusion appear simultaneously in matrix nanopores with mean diffusion coefficients equaling to 4.99×10^{-14} m²/s for pore of diameter smaller than 4 nm, and 9.09×10^{-9} m²/s for pores of diameter bigger than 4 nm. Some values of surface diffusion coefficient are listed in Table 8. The measured value from Yang et al. [55] is obviously lower than others. The reason may be that surface diffusion has a positive relation with pressure, and it was measured at low pressure condition in Yang et al. [55] comparing to other studies in Table 8.

As we can see from above introduction, the role of surface diffusion in shale gas transport has not reach an agreement up to date. Surface diffusion can analytically be modelled in different ways, and its contribution can also be studied in different pore size, pressure and temperature conditions. However, shale gas migration in formation is a complex fluid–solid system, including different pore types, pore shapes and pore size distribution, which need to be investigated in depth to quantify the significance of surface diffusion in shale gas transport process as well as its influence on shale gas production. Corresponding study need to be further investigated in the future.

12. Gas transporting model in shale nanopores

Bulk phase gas transport models and surface diffusion models in shale nanopores are reviewed and discussed in above sections. For gas transporting in nanopores, the total gas transporting flux is the summation of bulk phase gas transport and surface diffusion, where linear superposition of these two flux is taken. According to the different part of flux, superposition models of gas transporting in shale nanopores can be divided into three categories, as shown in Fig. 37. Fig. 37(a) considers double slippage in bulk phase, where the bulk phase flux need to be calculated by the method of M-2 or M-7 in Table 7. Fig. 37(b) considers single slippage in bulk phase, where the bulk phase flux is recommended to be calculated by the method of M-8. In this way, the calculation results of double slippage model and single slippage model achieve unification. Yin et al. [9] pointed out that surface diffusion is not only a movement of adsorbed gas, but also promotes the free gas flow in the middle of the pore space, as the yellow part shown in Fig. 37(c). Studies considering this part of gas movement are not common in current research, which need attention in further study.

13. Conclusion, discussion and outlook

The understanding of micro seepage mechanisms in shale gas nanopores is not only a critical and hot research topic, but also a key and practical issue to be determined for efficient exploitation of shale gas resources. Though much progress has been achieved in the last one or two decades, divergence and confusion on this subject are also transparent and in urgent need to be tackled. This review provides a comprehensive summarization and discussion on analytical modeling of gas transporting behaviors in shale gas reservoirs. The effects of geo-mechanical, gas sorption and real gas on gas flow behaviors in shale nanopores are considered, with different methods and approaches given and compared for each subject. Considering these effects, gas mean free path and Knudsen number range are calculated based on practical shale gas reservoir conditions. Slippage effect and Knudsen diffusion were modeled for bulk phase gas transport, with eight type of models classified from current studies. Results show that some models are inappro-

priate to describe gas flow behaviors, while other models can achieve unification, no matter they belong to double slippage model or single slippage model. Linear superposition of slippage flow flux and Knudsen diffusion flux cannot explain slippage effect and Knudsen diffusion simultaneously, but reconsiders the slippage effect and overestimates bulk phase gas transporting capacity. At last, different models of surface diffusion were also concluded and compared. From our review, the following conclusions and advices are given:

- (1) For slip flow, there are great discrepancies from different slip boundary conditions. Thus, these slip boundary conditions are recommended to be further evaluated according to the practical conditions of shale gas reservoirs, considering the fact that dense gas flow in confined shale porous media is quite different from that in other micro pores.
- (2) Surface diffusion modeling also has not achieved consensus currently, with the discrepancies from two type of models reaching 2 – 3 times order approximately, which needs particular emphasis and attention in further investigations. Lack of accurate, reliable and targeted experimental results is one of the reasons for the failure of consistency and selection among various models, since most experimental data in literature are either from other fields or not designed according to real shale gas reservoirs conditions. This is also true for the selection of numerous slip models for shale gas reservoirs.
- (3) Current analytical gas transporting models in shale gas reservoirs are mainly established based on single circular or rectangular shaped capillary, where heterogeneity and anisotropy characteristics of shale formations are not considered. Though some procedures could take porous media properties of porosity and tortuosity (ϕ/τ) into account, this kind of modification is rough with low accuracy. Besides, the strong heterogeneity and anisotropy characteristics of shale gas reservoirs cannot be processed by this simple procedure.
- (4) Shale gas production leads to pressure decrease in whole reservoir area, while gas output passes through multi-scaled flow paths. The variable pressure and multi-scale system require fully coupling of multiple fluid and reservoir physics, while current micro seepage models lack the coupling with larger scales as well as including full physical parameters.

Based on the above discussion, the following works need to be done in the future

- (1) Accurate and robust experimental design on slip flow and surface diffusion is in need to be conducted respectively to access more targeted experimental data, which helps to validate theoretical models and clarify micro seepage mechanisms.
- (2) More advanced and accurate gas transporting models considering variable cross section area, heterogeneity and anisotropy characteristics, phase behavior change, initial water effects *et al.* need to be developed according to practical reservoir conditions.
- (3) More accurate multi-scale, multi-field and multi-component gas transporting models based on practical shale gas reservoir conditions need to be developed.
- (4) Knudsen number is not the only flow similarity parameter between rarefied micro flow and dense shale gas flow. Thus, the study of shale gas flow characteristics cannot totally copy that of rarefied gas flows, which need to be conducted separately based on dense gas theory.

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Declaration of interest

None.

Appendix A. Derivation of $f(Kn)$ in paths with different shapes

(1) The derivation of $f(Kn)$ for gas flow in a microchannel (Fig. 22)

Ignoring gravitational body force and assuming steady state flow, the governing equation for Poiseuille flow confined in two paralleled plates can be written in the coordinate system shown in Fig. 22.

$$\frac{d^2 u}{dz^2} = \frac{1}{\mu} \frac{dp}{dx} \quad (A1)$$

where u is gas flow velocity in x direction.

According to the physical approach, the center symmetry condition need to be satisfied as

$$\left. \frac{du}{dz} \right|_{z=0} = 0 \quad (A2)$$

The second order slip boundary condition can be written as

$$u|_{z=h/2} = -C_1 \lambda \left(\frac{du}{dz} \right) \Big|_{z=h/2} - C_2 \lambda^2 \left(\frac{d^2 u}{dz^2} \right) \Big|_{z=h/2} \quad (A3)$$

Solving the mathematical model of Eqs. (A1), (A2) and (A3), the velocity distribution across the paralleled plates can be obtained as

$$u = -\frac{h^2}{8\mu} \frac{dp}{dx} \left[1 + 4C_1 Kn + 8C_2 Kn^2 - 4\left(\frac{z}{h}\right)^2 \right] \quad (A4)$$

According to velocity distribution, the average velocity can be solved as

$$u_{avg} = \frac{1}{h} \int_{-h/2}^{h/2} u dz = -\frac{h^2}{12\mu} \frac{dp}{dx} (1 + 6C_1 Kn + 12C_2 Kn^2) \quad (A5)$$

Clearly, the modification function $f(Kn)$ for two paralleled plates or a duct with rectangular cross section can be expressed as

$$f(Kn) = 1 + 6C_1 Kn + 12C_2 Kn^2 \quad (A6)$$

where C_1 and C_2 are model parameters and differ in different model, which can be seen in 7.2 Slip boundary conditions.

(2) The derivation of $f(Kn)$ for micro-tube with circular cross section (Fig. 23)

Similarly, the Poiseuille flow in a long micro-tube with uniform cross section can be described by the following equation taking the coordinate system shown in Fig. 23,

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{du}{dr} \right) = \frac{1}{\mu} \frac{dp}{dx} \quad (A7)$$

The second order slip boundary condition can be expressed as

$$u|_{r=R} = -C_1 \lambda \left(\frac{du}{dr} \right) \Big|_{r=R} - C_2 \lambda^2 \left(\frac{d^2 u}{dr^2} \right) \Big|_{r=R} \quad (A8)$$

Meanwhile, gas velocity in the micro-tube need to be a finite value

$$u|_{r=0} = \text{const} \quad (A9)$$

where const denotes a number with finite value.

Solving the mathematical model of Eqs. (A7), (A8) and (A9), we can obtain the velocity distribution in a long micro-tube as

$$u = -\frac{R^2}{4\mu} \frac{dp}{dx} \left[1 + 4C_1 Kn + 8C_2 Kn^2 - \left(\frac{r}{R}\right)^2 \right] \quad (A10)$$

Integrating velocity distribution of Eq. (A10) across the radius, we can obtain the volumetric flux in the micro-tube as

$$Q = \int_0^R 2\pi r u dr = -\frac{\pi R^4}{8\mu} \frac{dp}{dx} (1 + 8C_1 Kn + 16C_2 Kn^2) \quad (A11)$$

Therefore, the modification function for gas flow in a long micro-tube can be written as

$$f(Kn) = 1 + 8C_1 Kn + 16C_2 Kn^2 \quad (A12)$$

Note that, the characteristic length was taken as pore diameter when defining Knudsen number in Eq. (73). If the pore radius is taken, the modification should be rewritten as

$$f(Kn) = 1 + 4C_1 Kn + 4C_2 Kn^2 \quad (A13)$$

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